HARAMAYA UNIVERSITY
COLLEGE OF COMPUTING AND INFORMATICS
DEPARTMENT OF SOFTWARE ENGINEERING

End-to-End Supply Chain Management System

A Senior Project Software Requirement Specification Submitted for Partial Fulfillment of the Requirements for BSc. Degree in Software Engineering

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Declaration Sheet

The Software Requirements Specification (SRS) for the End-to-End Supply Chain Management System presented in this document is the original work of the group members whose names are listed below, completed under the supervision of Ms. Meti Dejene, without any unauthorized external assistance.

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I witnessed that this document was done according to the guidelines under my supervision. This final document incorporates my comments.

Mety

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Abbreviations and Acronyms

AI – Artificial Intelligence

API – Application Programming Interface

CRUD – Create, Read, Update, Delete

CSV – Comma-Separated Values

DB – Database

DBMS – Database Management System

ERD – Entity Relationship Diagram

FK – Foreign Key

GUI – Graphical User Interface

HTTP – Hypertext Transfer Protocol

HTTPS – Hypertext Transfer Protocol Secure

ID – Identifier

JSON – JavaScript Object Notation

NFR – Non-Functional Requirement

PK – Primary Key

PO – Purchase Order

RBAC – Role-Based Access Control

RDBMS – Relational Database Management System

REST – Representational State Transfer

SCM – Supply Chain Management

SDD – Software Design Document

SRS – Software Requirements Specification

SQL – Structured Query Language

UI – User Interface

UML – Unified Modeling Language

URL – Uniform Resource Locator

UX – User Experience

WM – Warehouse Manager

WWW – World Wide Web

CHAPTER ONE: INTRODUCTION

1.1 Document Scope

This SDD describes the architectural and detailed design of the **Web-Based End-to-End SCM System.** The document translates the requirements specified in the SRS into a structured design that guides system implementation.

The SDD presents the system architecture, subsystem decomposition, data design, user interface design, and requirement traceability. It serves as a blueprint for developers and provides evaluators with a clear understanding of how the system requirements are realized through design decisions.

This document is organized as follows:

I. Chapter One introduces the purpose, scope, conventions, and intended audience of the document.
II. Chapter Two presents the system architectural design using the 4+1 view model.
III. Chapter Three describes logical and physical data design.
IV. Chapter Four explains the human interface design.
V. Chapter Five provides the requirement traceability matrix.

1.2 Document Purpose

The purpose of this SDD is to define a clear and detailed design for the **Web-Based End-to-End SCM System** based on the approved SRS.

Specifically, this document aims to:

- ✓ Translate functional and non-functional requirements into a concrete system architecture
- ✓ Provide a technical reference for system implementation
- ✓ Support academic evaluation by demonstrating sound software design principles
- ✓ Ensure traceability between requirements and design components

- ✓ Serve as a reference for future system enhancement and maintenance

1.3 Document Conventions

The following conventions are used throughout this document:

- ✓ Section numbering follows a hierarchical decimal format (e.g., 2.4.1)
- ✓ UML diagrams are used to illustrate architectural and design views
- ✓ Bold text is used for headings and key terms
- ✓ Acronyms and abbreviations are defined in the "Definitions, Abbreviations, and Acronyms" section
- ✓ Technical terms follow standard software engineering terminology
- ✓ All diagrams conform to UML standards and are compatible with tools such as draw.io.

1.4 Intended Audience

This document is intended for the following audiences:

- ✧ **Development Team**: To guide system implementation and integration
- ✧ **Project Advisor**: To evaluate architectural soundness and design decisions
- ✧ **University Examiners**: To assess compliance with academic and technical standards

CHAPTER TWO: SYSTEM ARCHTECTURAL DESIGN

2.1 System Overview

The **Web-Based End-to-End SCM** is a centralized web application designed to automate and streamline supply chain operations between suppliers and retailers while ensuring administrative oversight without business interference.

The system enables direct interaction between suppliers and retailers for procurement, product discovery, order, placement, inventory, management, delivery tracking, and performance evaluation. Administrators function solely as platform controllers responsible for business license verification, system monitoring, and dispute resolution.

The system follows an end-to-end workflow, beginning with user registration and verification, followed by procurement and PO creation, product listing, supplier discovery, order processing, delivery execution, and performance evaluation. All activities are performed digitally to improve transparency, efficiency, and traceability across the supply chain.

2.1.1 Key Objectives of the System

- Replace manual and fragmented supply chain processes with a unified digital platform.
- Enable structured procurement through formal PO's.
- Support supplier-owned warehouse and inventory management.
- Facilitate vehicle rental coordination between suppliers and car rental agencies.
- Support real-time order and delivery tracking.
- Ensure regulatory compliance through business license verification.
- Prevent administrative interference in supplier--retailer transactions.

2.1.2 Overall System Workflow

The high-level workflow of the system is summarized as follows:

1. **User Registration and Verification**
   Suppliers, retailers, delivery agents, and car rental agencies register by submitting

business details and licenses.

Administrators verify license validity before account activation.

2. **Product and Supplier Discovery**

   Suppliers list products and manage inventory.

   Retailers search and filter suppliers based on product type, quantity, price, location, and rating.

3. **Procurement & PO Creation**

   Retailers issue formal POs to selected suppliers.

   Suppliers accept or reject POs. Accepted POs are automatically converted into executable orders.

4. **Order Processing**

   Orders are generated from approved POs.

   Suppliers manage order fulfillment and inventory is updated automatically.

   Order splitting is supported when a single supplier cannot fulfill the full quantity.

5. **Delivery and Logistics Coordination**

   Suppliers arrange deliveries using owned vehicles or through rental requests to car rental agencies.

   Delivery agents are assigned and update delivery status throughout the process.

   Delivery status is tracked by retailers.

6. **Delivery Confirmation and Rating**

   Retailers confirm delivery receipt.

   Suppliers are rated based on delivery performance and product quality.

7. **Administrative Oversight**

   Administrators monitor system activity, verify licenses, and resolve disputes without participating in transactions.

2.1.3 System Context

The system interacts with five primary external actors:

1. **Supplier** -- Manages products, inventory, POs, orders, and deliveries; requests vehicle rentals.

2. **Retailer** -- Searches products, issues PO, places orders, tracks deliveries, and provides ratings.
3. **Delivery Agent** -- Updates delivery status and confirms handover; does not access transactional or financial data.
4. **Car Rental Agency** -- Registers vehicles, manages availability, and responds to rental requests from suppliers.
5. **Administrator** -- Verifies licenses, monitors system integrity, and resolves disputes.
6. **Warehouse manager**-- Operates under supplier authority; prepares goods, manages inventory allocation, and releases items for delivery

A high-level **system context diagram** is used to illustrate interactions between these actors and the core system modules.

2.2 Design Goals

The design of the **Web-Based End-to-End SCM System** aims to satisfy the non-functional requirements specified in the SRS. The system architecture and design decisions focus on performance, usability, security, reliability, scalability, and maintainability to ensure effective operation within the academic and business context.

Performance

The system is designed to provide acceptable response times and throughput under normal usage conditions. A web-based, modular architecture is adopted to reduce processing overhead and ensure efficient handling of user requests. Database indexing, optimized queries and separation of concerns between system modules are used to support timely procurement, product searches, order processing, and report generation.

This design ensures that key user interactions such as dashboard loading, PO creation, product searching, and order placement are completed within the response time limits defined in the SRS.

Usability

Usability is a primary design goal due to the varied technical skill levels of system users. The system follows a simple, role-based user interface design that presents only relevant features to each user type (Supplier, Retailer, Administrator, Delivery Agent, and Car Rental Agency). Clear navigation, minimal input steps, and informative feedback messages are incorporated to reduce user errors and learning time.

The design ensures that common tasks such as procurement, product listing, order placement, vehicle rental coordination, and delivery tracking can be completed with minimal effort and without requiring advanced technical knowledge.

Security

Security is addressed through role-based access control, authentication mechanisms, and secure data handling practices. The system design restricts access to features based on user roles to prevent unauthorized actions. Sensitive data such as user credentials, business documents, and payment information are protected through secure storage and encrypted communication channels.

Audit logging is included in the design to track critical system activities such as user registration approval, PO approval, order processing, and rental transactions, supporting accountability and dispute resolution.

Reliability and Availability

The system is designed to operate reliably during normal business hours with minimal downtime. Error handling mechanisms and data validation rules are incorporated to prevent system failures and data inconsistencies. The design ensures that failures are handled gracefully without loss of critical data, particularly in procurement and vehicle rental workflows.

Backup and recovery considerations are included to support system reliability and continuity in case of unexpected failures.

Scalability

Although the system is developed within an academic environment, the design supports future scalability. A modular architecture allows individual subsystems such as procurement management, order processing, inventory management, vehicle rental coordination, and delivery tracking to be extended or optimized independently. The system design supports increased numbers of users, POs, products, vehicles, and transactions without requiring major architectural changes.

Maintainability

Maintainability is achieved through modular subsystem decomposition, clear separation of responsibilities, and consistent design standards. Each subsystem—including newly introduced procurement and vehicle rental modules is designed to be independently maintainable and testable, making future enhancements and bug fixes easier to implement.

2.3 Subsystem Decomposition

The **Web-Based End-to-End SCM System** is decomposed into a set of independent yet interrelated subsystems. Each subsystem represents a logical grouping of functionality that can operate independently while collaborating with other subsystems through well-defined interfaces.

This decomposition improves system modularity, maintainability, scalability, and ease of development.

2.3.1 Identified Subsystems

The system is divided into the following major subsystems:

1. User Management Subsystem

Description:
This subsystem manages user registration, authentication, authorization, and business license verification for all system roles.

Responsibilities:

- ✧ Register suppliers, retailers, administrators, delivery agents, car rental agencies, and warehouse managers.
- ✧ Handle login and logout functionality
- ✧ Enforce role-based access control
- ✧ Manage business license submission and approval
- ✧ Maintain audit logs for registration and approval activities

Actors: Administrator, Supplier, Retailer, Warehouse Manager, Delivery Agent, Car Rental Agency

2. Procurement & PO Management Subsystem

Description:
This subsystem manages the formal procurement process through POs, ensuring structured approval before order creation.

Responsibilities:

- ✧ Create and manage POs.
- ✧ Handle PO acceptance or rejection by suppliers
- ✧ Convert approved POs into executable orders
- ✧ Maintain PO history for auditing and reporting
- ✧ Track PO status throughout the procurement lifecycle

Actors: Retailer, Supplier

3. Product and Catalog Management Subsystem

Description:
This subsystem allows suppliers to manage product listings and organize product information.

Responsibilities:

- Create, update, and delete product listings
- Maintain product categories and attributes
- Associate products with supplier-owned warehouses
- Provide product data for search and filtering

Actors: Supplier

4. Supplier Discovery and Search Subsystem

Description:
This subsystem enables retailers to search and filter suppliers and products based on business criteria.

Responsibilities:

- Search products and suppliers
- Apply filters such as price, quantity, location, and rating
- Display supplier performance information
- Support supplier selection for procurement

Actors: Retailer

5. Order Processing Subsystem

Description:
This subsystem manages the order lifecycle, converting approved POs into executable orders and handling order fulfillment.

Responsibilities:

- Convert approved POs into orders
- Handle order status updates
- Support partial fulfillment and order splitting
- Track order history

- ✧ Prevent administrative interference in transactions

Actors: Retailer, Supplier

6. Inventory and Warehouse Management Subsystem

Description:
This subsystem manages inventory levels within supplier-owned warehouses.

Responsibilities:

- ✧ Track inventory quantities
- ✧ Automatically update stock levels after order confirmation
- ✧ Generate low-stock alerts
- ✧ Maintain inventory history records

Actors: Supplier, Warehouse Manager

7. Delivery and Logistics Management Subsystem

Description:
This subsystem handles delivery execution and tracking, coordinating with the Vehicle Rental Management Subsystem when external vehicles are required.

Responsibilities:

- ✧ Manage delivery status updates
- ✧ Track delivery stages from dispatch to completion
- ✧ Provide estimated delivery timelines
- ✧ Support delivery confirmation and proof collection
- ✧ Interact with Vehicle Rental Management for vehicle assignment

Actors: Supplier, Retailer, Delivery Agent

8. Vehicle Rental Management Subsystem

Description:

This subsystem manages vehicle rental requests from suppliers to car rental agencies and coordinates vehicle assignment to deliveries.

Responsibilities:

- ✧ Register and manage vehicle details by car rental agencies
- ✧ Handle rental requests from suppliers
- ✧ Process acceptance or rejection of rental requests
- ✧ Assign vehicles to specific deliveries
- ✧ Track vehicle usage and availability

Actors: Supplier, Car Rental Agency

9. Warehouse Operations Management Subsystem

Description:

This subsystem manages warehouse operations including inventory handling, order preparation, and stock management under supplier authority.

Responsibilities:

- ✧ Manage inventory picking and packing operations
- ✧ Update stock levels during order fulfillment
- ✧ Track warehouse location assignments
- ✧ Generate restock requests to suppliers
- ✧ Maintain warehouse activity logs

Actors: Warehouse Manager, Supplier

10. Rating and Feedback Subsystem

Description:

This subsystem allows retailers to evaluate supplier performance after delivery completion.

Responsibilities:

- ✧ Collect ratings and feedback
- ✧ Calculate average supplier ratings
- ✧ Display performance metrics
- ✧ Support feedback responses by suppliers

Actors: Retailer, Supplier

11. Reporting and Analytics Subsystem

Description:

This subsystem provides reporting and analytics for all system users.

Responsibilities:

- ✧ Generate role-based reports
- ✧ Provide system-wide analytics for administrators
- ✧ Support data export in multiple formats
- ✧ Display business performance insights

Actors: Administrator, Supplier, Retailer

12. Administrative Oversight Subsystem

Description:

This subsystem supports administrative monitoring and compliance management without business interference.

Responsibilities:

- ✧ Verify business licenses
- ✧ Monitor system activity and audit logs
- ✧ Handle dispute resolution
- ✧ Generate compliance and performance reports

Actors: Administrator

13. Notification and Alert Management Subsystem

Description:

This subsystem is responsible for generating, managing, and delivering internal system notifications and alerts triggered by key system events.

Responsibilities:

- ✧ Generate notifications for key system events including PO creation, order updates, delivery status changes, rental requests, and payment confirmations
- ✧ Deliver role-based notifications to users
- ✧ Maintain notification status (read/unread)
- ✧ Store notifications for historical and audit purposes

Actors: Supplier, Retailer, Administrator, Delivery Agent, Car Rental Agency, Warehouse Manager

2.3.2 Subsystem Relationships

- The system subsystems interact through well-defined internal interfaces to ensure seamless data flow while maintaining separation of concerns. Key subsystem interactions are as follows:
- **Procurement and PO Management Subsystem**
  - Receives approved Purchase Orders (POs) from **Retailers** for conversion into executable system-generated Orders.
  - Supports **PO splitting** when a single PO must be fulfilled through multiple orders due to inventory or delivery constraints.

- **Order Processing Subsystem**
    - Coordinates with the **Inventory and Warehouse Management Subsystem** to reserve inventory upon order confirmation.
    - Notifies the **Warehouse Operations Management Subsystem** of orders requiring preparation.
    - Ensures inventory quantities are permanently deducted only after successful dispatch to prevent over-allocation and maintain stock consistency.
- **Warehouse Operations Management Subsystem**
    - Tracks stock movements in coordination with the Inventory subsystem.
    - Prepares orders for dispatch in collaboration with the Delivery and Logistics Management Subsystem.
    - Reports inventory changes and order preparation status to suppliers via **authenticated user access**, enforced by the **User Management Subsystem**, ensuring accountability.
- **Delivery and Logistics Management Subsystem**
    - Updates order status within the **Order Processing Subsystem**.
    - Coordinates with the **Vehicle Rental Management Subsystem** when supplier delivery capacity is unavailable internally.
- **Vehicle Rental Management Subsystem (VRMS)**
    - Automatically checks **supplier-owned vehicle availability** for the requested delivery date before PO acceptance.
    - If no internal vehicles are available, queries external **Car Rental Agencies (CRAs)** via API/web service for vehicle availability.
    - Vehicle assignment (internal or rented) must be confirmed before the supplier can accept the PO.
    - Coordinates with the Delivery and Logistics Management Subsystem to schedule deliveries and tracks vehicle usage.
    - Notifies the Warehouse Operations Management Subsystem of vehicle availability to support order preparation scheduling.
- **Rating and Feedback Subsystem**

- Consumes data from completed orders supplied by the **Order Processing Subsystem**.
- Activated only after an order is successfully delivered to ensure feedback is collected exclusively for completed transactions.

- **Administrative Oversight Subsystem**
  - Accesses audit logs and reporting data in a read-only manner.
  - Monitors activities across operational subsystems without interfering with transactional workflows.
- **Notification and Alert Management Subsystem**
  - Listens to key events generated by multiple subsystems—including Procurement, Order Processing, Inventory Management, Warehouse Operations, Delivery and Logistics, and Administrative Oversight.
  - Delivers **role-based internal notifications** while maintaining notification history for auditing, without affecting transactional processing.
- **User Management Subsystem**
  - All subsystems requiring authentication and authorization interact with this subsystem to **validate user credentials** and enforce access control policies before performing sensitive operations.

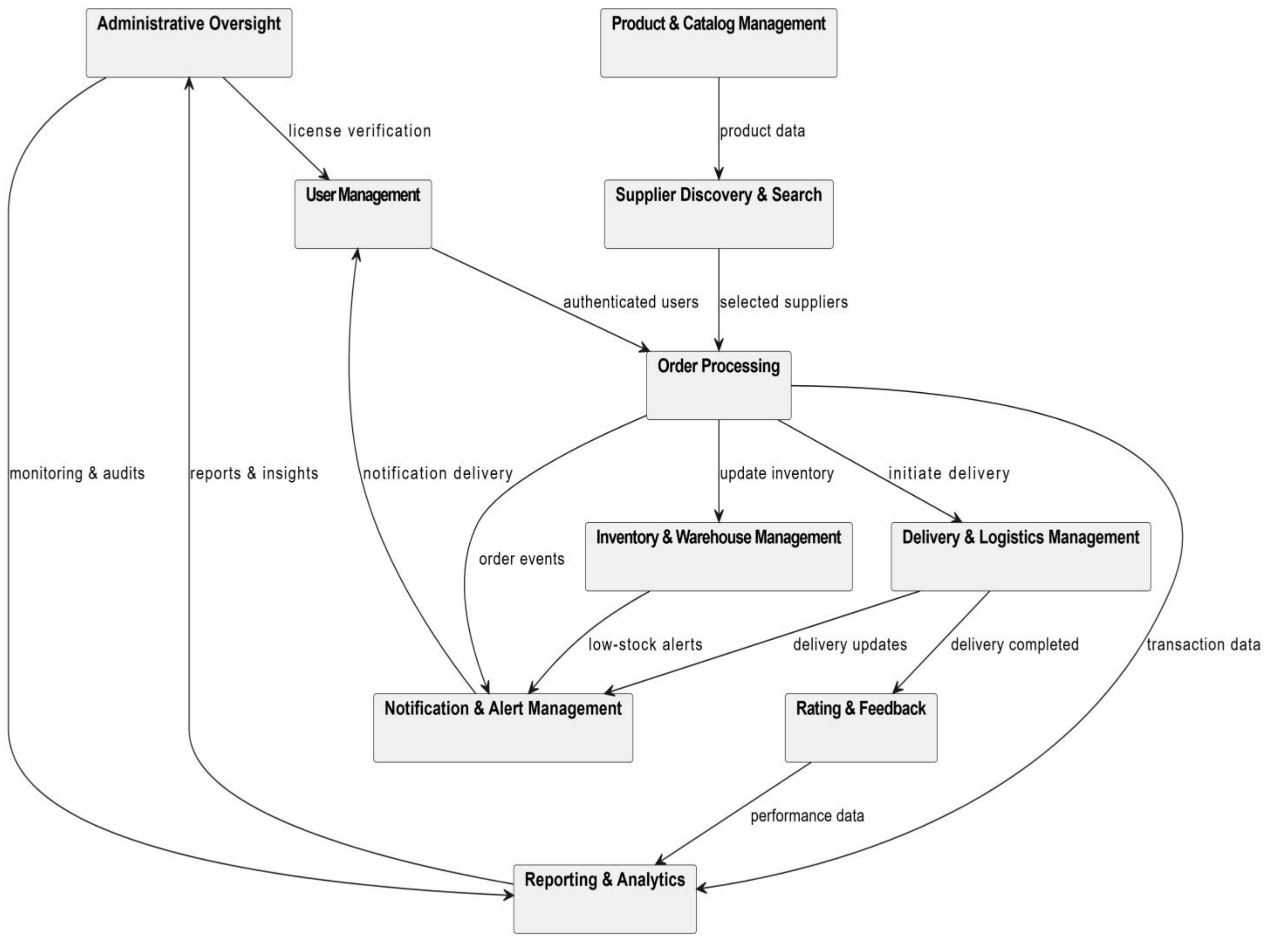

Figure 2.3 – Subsystem Decomposition Package Diagram

2.4 4+1 Architecture View Model

2.4.1 Logical View

The logical view describes the static structure of the Web-Based End-to-End SCM System and focuses on the system's functional elements and their relationships. This view illustrates how the system is organized to fulfill user requirements without considering implementation or deployment details.

The logical view is derived from the use case model and the core entities defined in the SRS. It identifies the main system classes and their interactions that support end-user functionality.

2.4.1.1 Key Logical Components

The system's logical structure is centered on the following core entities:

- **User:** Represents all system users, specialized into Administrator, Supplier, Retailer, Warehouse Manager, Delivery Agent, and Car Rental Agency roles.
- **PurchaseOrder (PO):** Represents procurement documents created by retailers and approved by suppliers before order creation.
- **Product:** Items listed by suppliers, including product details, pricing, and availability.
- **Warehouse:** Supplier-owned storage locations associated with inventory.
- **Inventory:** Tracks stock levels of products within warehouses.
- **Order:** Executable orders generated from approved POs.
- **OrderItem:** Individual product entries within an order.
- **VehicleRental:** Vehicle rental requests for delivery purposes.
- **Delivery:** Delivery process associated with an Order.
- **Rating:** Performance evaluations provided by retailers after delivery completion.
- **Report:** System-generated analytical and compliance reports.
- **PaymentConfirmation:** Records external proof of commission payments; system does not process actual transactions.
- **Commission:** Records commission calculations linked to orders.

2.4.1.2 Logical Relationships

- A Supplier manages one or more Warehouse Managers and lists multiple Products.
- A Warehouse Manager operates under a Supplier's authority and manages Inventory within assigned Warehouses.
- A Retailer creates one or more PurchaseOrders.
- A PurchaseOrder may be converted into one or more Orders (1:N) to support partial fulfillment or delivery constraints while maintaining traceability to the original PO.
- An Order consists of one or more OrderItems**,** each referencing a Product.

- Inventory is associated with Products and Warehouses to track available stock. Reserved quantities are held at order confirmation and permanently deducted after successful dispatch.
- Each Order is associated with a Delivery.
- A Delivery may be associated with a VehicleRental (1:1); vehicle rental is optional and triggered only when delivery capacity requires it.
- A Retailer may provide a Rating after order delivery; feedback is allowed only for completed transactions.
- Administrators access Reports and audit data in read-only mode without modifying transactional entities.

This logical organization ensures clear separation of responsibilities while supporting all required system functionalities.

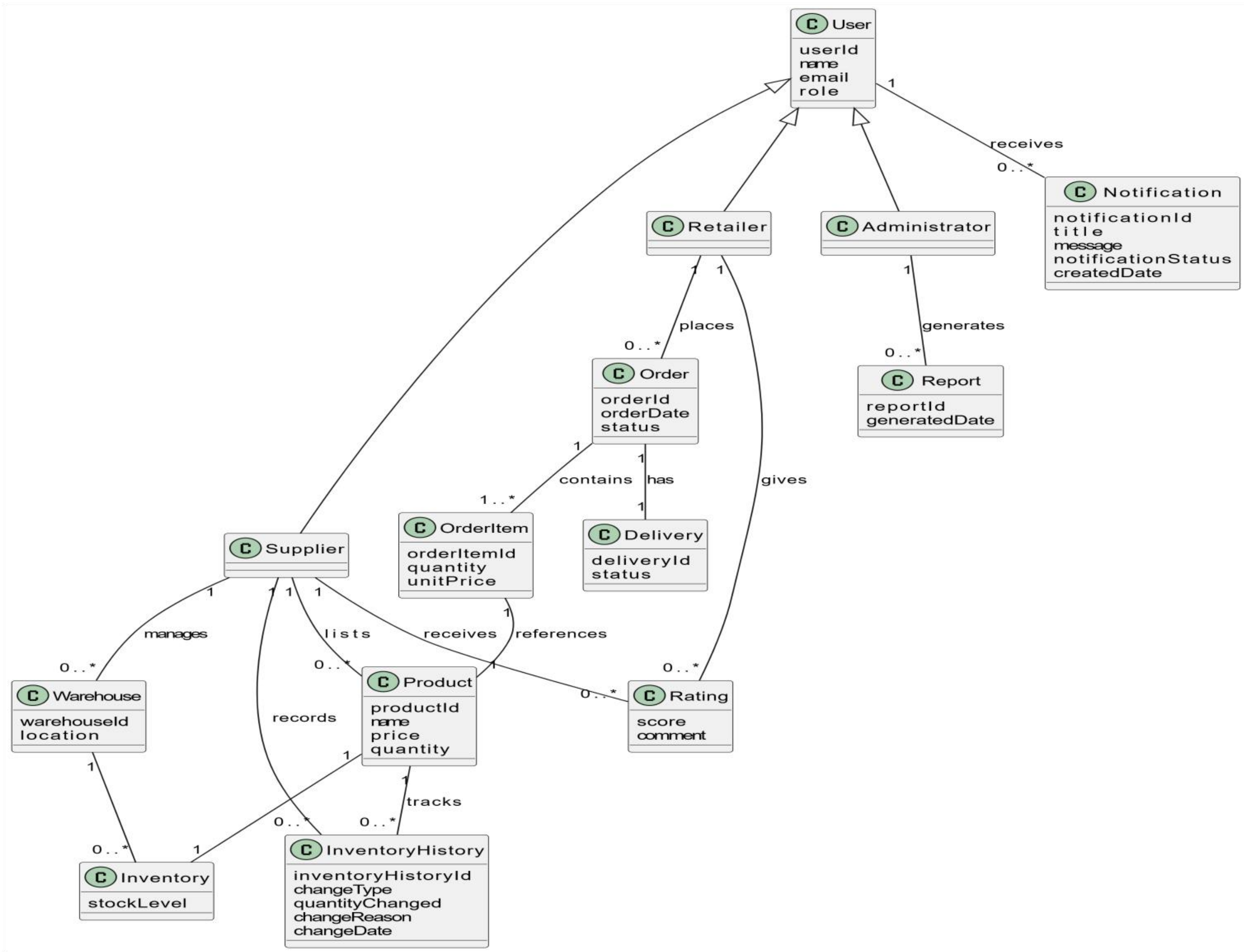

Figure 2.4.1 – Logical View Class Diagram

2.4.2 Process View

The process view describes the dynamic behavior of the Web-Based End-to-End SCM System and focuses on how system components interact during execution. It illustrates the flow of control, communication between subsystems, and the sequence of operations performed to fulfill user requests.

This view is especially important for understanding user registration, procurement, order processing, vehicle rental coordination, and delivery tracking, which are central to the system's functionality.

The process view is represented using **sequence diagrams** and **activity diagrams** to model runtime interactions and workflows.

2.4.2.1 Key System Processes

The major system processes include:

1. User registration and license verification (including all user types)
2. Procurement and PO creation and approval
3. Product discovery and supplier selection
4. Vehicle rental request and assignment
5. Order placement and confirmation
6. Order splitting and inventory updates
7. Delivery tracking and confirmation
8. Supplier rating and feedback (after successful delivery)

2.4.2.2 Process 1: User Registration and Verification

This process illustrates how suppliers, retailers, delivery agents, and car rental agencies register and how administrators verify business licenses before account activation.

Process Description:

1. **User Registration Submission**
   - User submits registration details, including personal/company information and business license (based on role).
   - CRA users submit additional information related to vehicles they manage (vehicle type, capacity, availability schedule).
2. **Pending State Storage**

- System stores registration data in a **pending state** until administrative verification is completed.

3. **Administrator Verification**
   - Administrator reviews submitted registration information and verifies business license validity.
   - For CRA users, the administrator also verifies vehicle registration and fleet details.
4. **Account Approval or Rejection**
   - If verification is successful, the account is **approved**; otherwise, it is **rejected**.
   - System notifies the user of the outcome.
5. **System Access Granted**
   - Approved users gain access to the system with permissions based on their role:
     - **Suppliers:** Manage products, accept POs, schedule deliveries.
     - **Retailers:** Create POs, track deliveries, rate suppliers.
     - **Delivery Agents:** View assigned deliveries and update delivery status.
     - **CRAs:** Manage vehicle availability, approve rental requests, and track vehicle assignments.

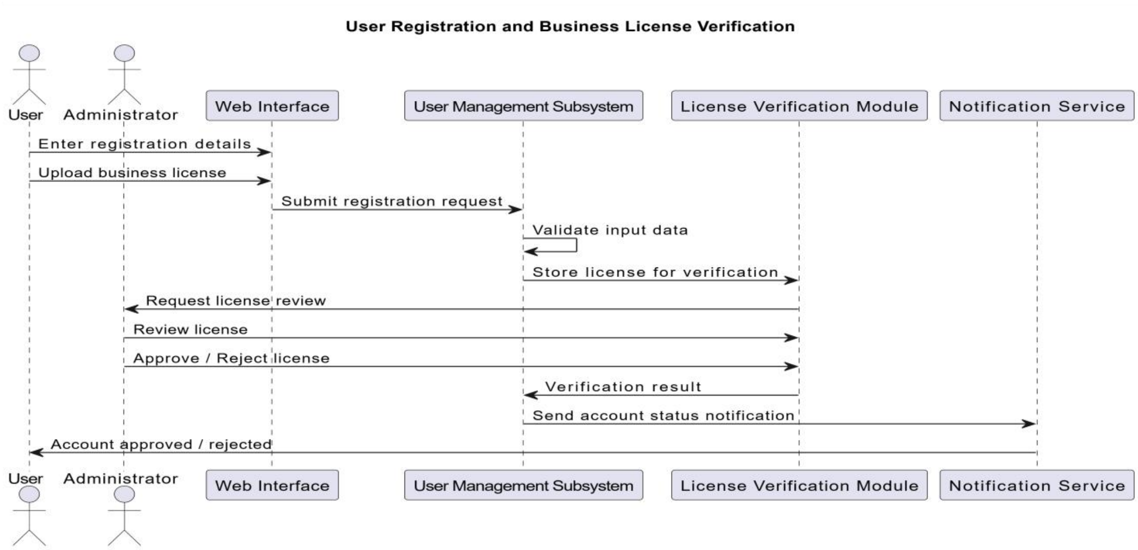

Figure 2.4.2(a) – User Registration and Business License Verification Sequence Diagram

2.4.2.3 Process 2: Procurement and Purchase Order Workflow

This process illustrates how retailers create POs and how suppliers approve them before order creation.

Process Description:

1. Retailer creates a PO with selected products
2. System stores PO with "Issued" status
3. Supplier reviews and accepts/rejects PO
4. If accepted, system converts PO to executable order
5. If rejected, retailer is notified and may modify/reissue PO

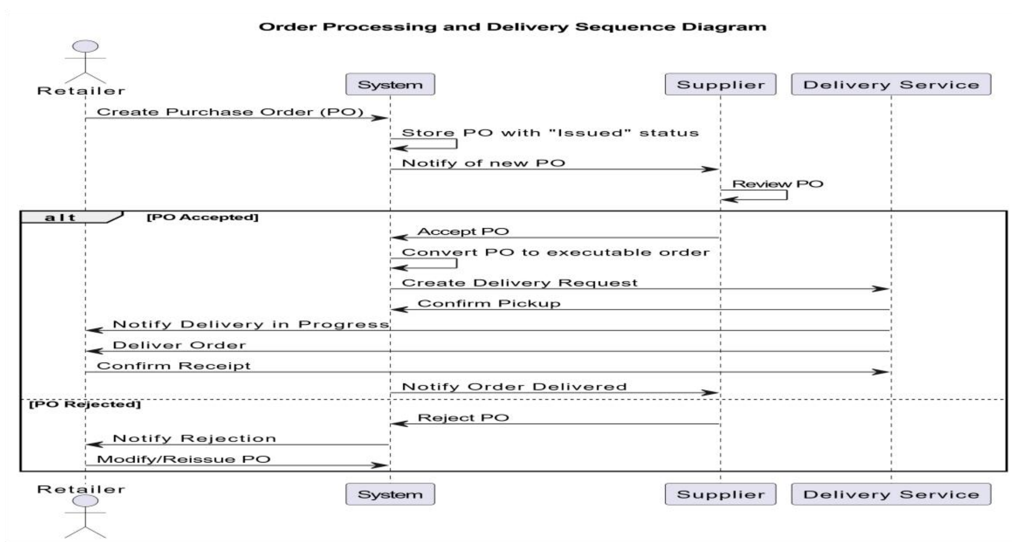

Figure 2.4.2(b) – Order Processing and Delivery Sequence Diagram

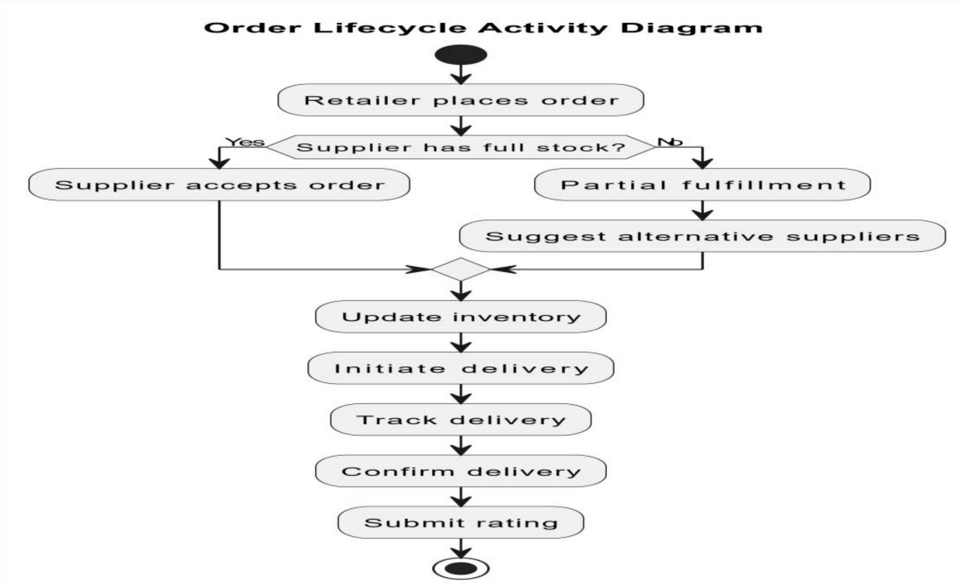

Figure 2.4.2(c) – Order Lifecycle Activity Diagram

2.4.2.4 Process 3: Vehicle Rental Request Workflow

This process illustrates how suppliers request vehicles from car rental agencies for delivery purposes.

Process Description:

1. **PO Submission**
    - Retailer submits a Purchase Order (PO) with a requested delivery date.
2. **Vehicle Availability Check**
    - The Vehicle Rental Management Subsystem automatically checks for **supplier-owned vehicles** available on the requested delivery date.
    - **If internal vehicles are available:**
        - Supplier may accept the PO.
        - Delivery is scheduled using supplier-owned vehicles.
    - **If no internal vehicles are available:**
        - System queries **external Car Rental Agencies (CRAs)** for available vehicles on the requested date.

- Supplier selects a CRA from available options.
- CRA approves or rejects the rental request.
- **Only after vehicle assignment** (rented) can the supplier accept the PO.

3. **PO Acceptance**
    - Supplier accepts the PO once a delivery vehicle (internal or rented) is confirmed.
4. **Delivery Execution**
    - Delivery agent executes the delivery using the assigned vehicle.
    - System tracks delivery progress, updates order status, and monitors vehicle usage until completion.

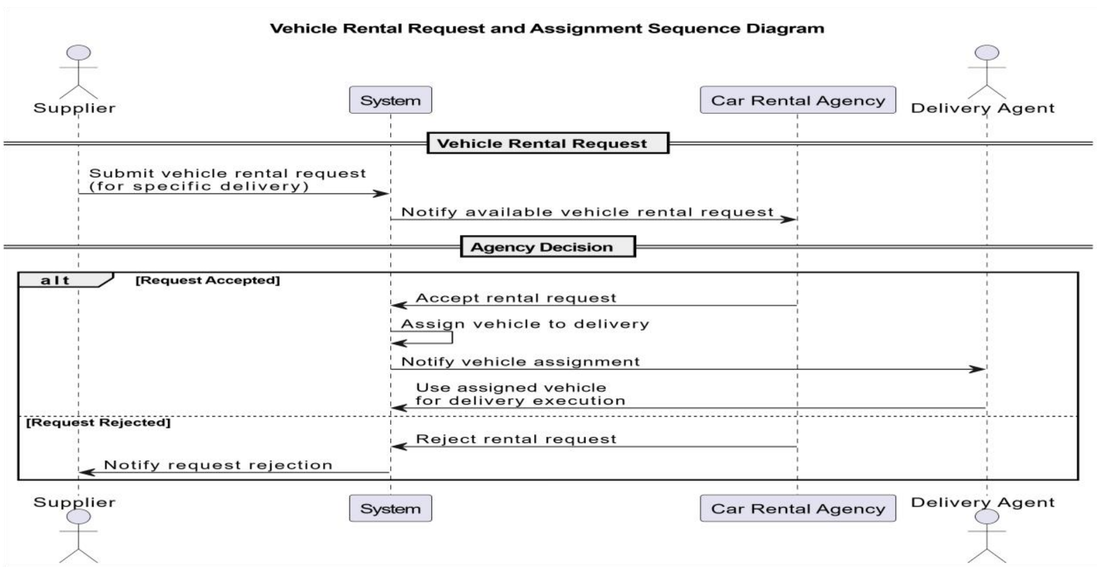

Figure 2.4.2(d) Vehicle Rental Request and Assignment Sequence Diagram

2.4.2.5 Process 4: Order Placement and Fulfillment

This process represents the core runtime behavior of the system, now integrated with procurement.

Process Description:

1. Retailer searches products and selects supplier
2. Retailer creates PO
3. Supplier reviews and accepts PO

4. System generates Order from approved PO
5. Inventory is reserved at order confirmation and deducted after dispatch.
6. Delivery process is initiated (optional VehicleRental if required)
7. Retailer tracks delivery
8. Retailer confirms delivery
9. Retailer rates supplier

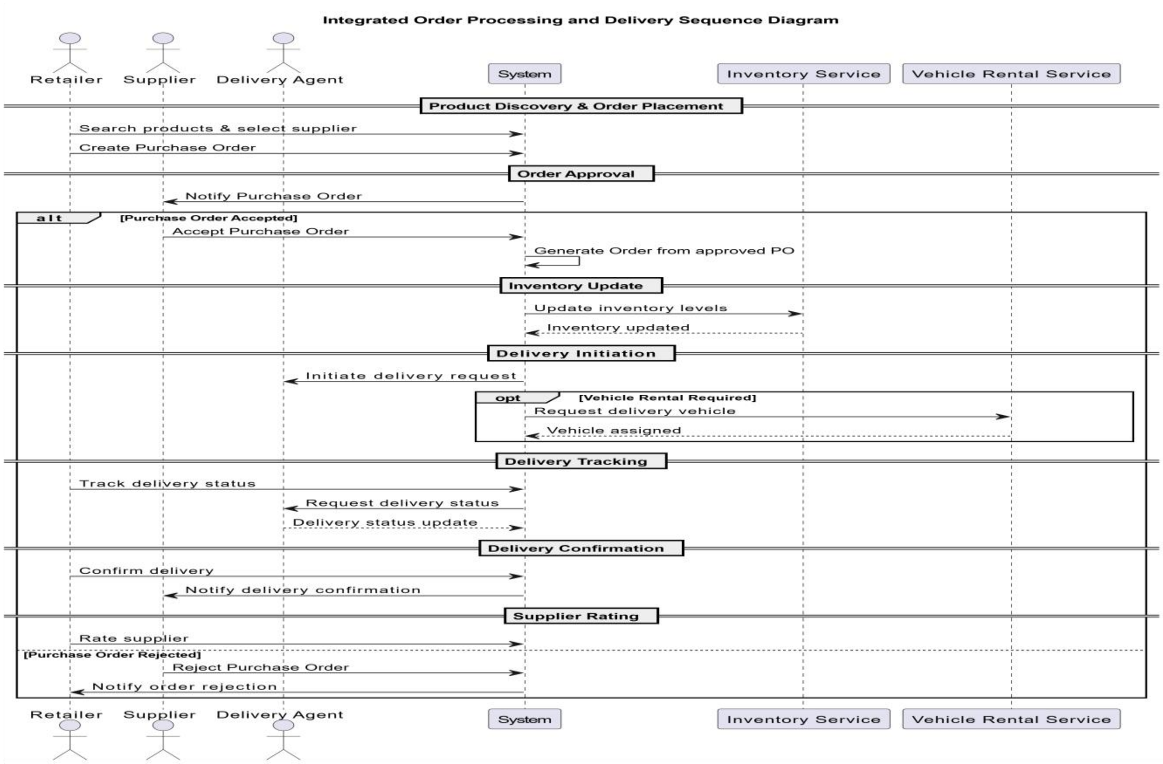

Figure 2.4.2(e) Integrated Order Processing and Delivery Sequence Diagram

2.4.3 Development View

The development view describes the software organization of the Web-Based End-to-End SCM System from an implementation perspective. It focuses on the major software components, their responsibilities, and the dependencies between them.

This view supports software development, maintenance, and future enhancement by clearly defining system modules and their interactions. The system follows a modular, layered architecture, where each component encapsulates a specific set of responsibilities.

2.4.3.1 Major System Components

The system is organized into the following key components:

1. Presentation Layer

Is Responsible for user interaction through a web-based interface.

- ✓ User dashboards (Supplier, Retailer, Administrator, Delivery Agent, Car Rental Agency)
- ✓ Forms for registration, procurement, product management, ordering, and delivery tracking
- ✓ Input validation and user feedback

2. Application / Business Logic Layer

It contains the core logic that enforces business rules and workflows.

- ✓ User Management Component
- ✓ Procurement Management Component
- ✓ Product & Catalog Management Component
- ✓ Supplier Discovery & Search Component
- ✓ Order Processing Component
- ✓ Inventory Management Component
- ✓ Delivery Management Component
- ✓ Vehicle Rental Management Component
- ✓ Rating & Feedback Component
- ✓ Reporting & Analytics Component
- ✓ Notification & Alert Management Component
- ✓ Notification Repository
- ✓ Inventory History Repository

3. Data Access Layer

Handles interaction with the database and data persistence.

- ✓ User Repository

- ✓ PurchaseOrder Repository
- ✓ Product Repository
- ✓ Order Repository
- ✓ Inventory Repository
- ✓ VehicleRental Repository
- ✓ Delivery Repository
- ✓ Rating Repository

2.4.3.2 Component Interaction Overview

- ✓ The Presentation Layer communicates with the Business Logic Layer through service interfaces.
- ✓ The Business Logic Layer enforces business rules and coordinates workflows.
- ✓ The Data Access Layer provides persistent storage and retrieval of system data.
- ✓ The Notification & Alert Management Component listens to events from procurement, order, delivery, vehicle rental, inventory, and administrative components and triggers notification delivery through internal services while persisting notification state.

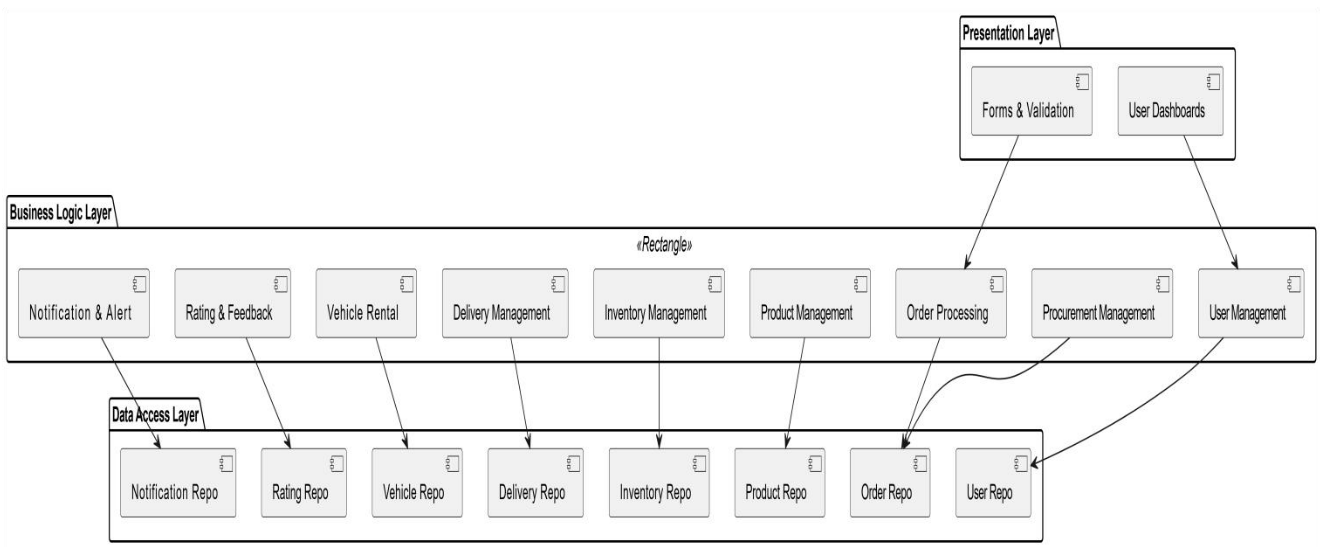

Figure 2.4.3 Development View Component Diagram

2.4.4 Physical View

The physical view describes the deployment of the Web-Based End-to-End SCM System on hardware nodes and network infrastructure. It defines how software components are mapped to servers, databases, and clients, ensuring scalability, reliability, and performance.

This view ensures that all subsystems and components from the Development and Logical Views can function effectively in a real environment.

2.4.4.1 Deployment Architecture

The system uses three**-tier web architecture**:

1. Client Tier (Web Browser)

- ✧ Web browsers used by Suppliers, Retailers, Administrators, Delivery Agents, and Car Rental Agencies
- ✧ Interacts with the system through HTTP/HTTPS
- ✧ Supports role-based dashboards and forms

2. Application Tier (Web Server)

- ✧ Hosts the Web Application (Presentation + Business Logic layers)
- ✧ Processes user requests
- ✧ Enforces business rules including procurement and vehicle rental workflows
- ✧ Handles sessions and authentication

3. Data Tier (Database Server)

- ✧ Stores persistent data (users, PO, products, orders, inventory, vehicle rentals, ratings)
- ✧ Supports transactions, queries, and backup
- ✧ Provides high availability and security

4. Supporting Services

- ✧ Internal Notification Service: Manages in-system dashboard alerts and notifications for account status, PO approvals, order updates, rental requests, and delivery notifications. It's initially implemented as an internal module within the Application Tier.
- ✧ File Storage Service: Stores business licenses, delivery proofs, vehicle documents, and other uploaded files.

Physical Nodes and Deployment Mapping

Node	Software Components	Responsibilities
Client (Browser)	Web UI	User interaction, input, and display
Web Server	Web Application (Business Logic + Presentation Layer)	Process requests, enforce business rules, generate and manage internal notifications, interact with DB
Database Server	DBMS + Data Access Layer	Persistent storage, transaction processing, store notification records
File Server	File Storage Service	Stores licenses, proofs, documents, vehicle records
Node	Software Components	Responsibilities

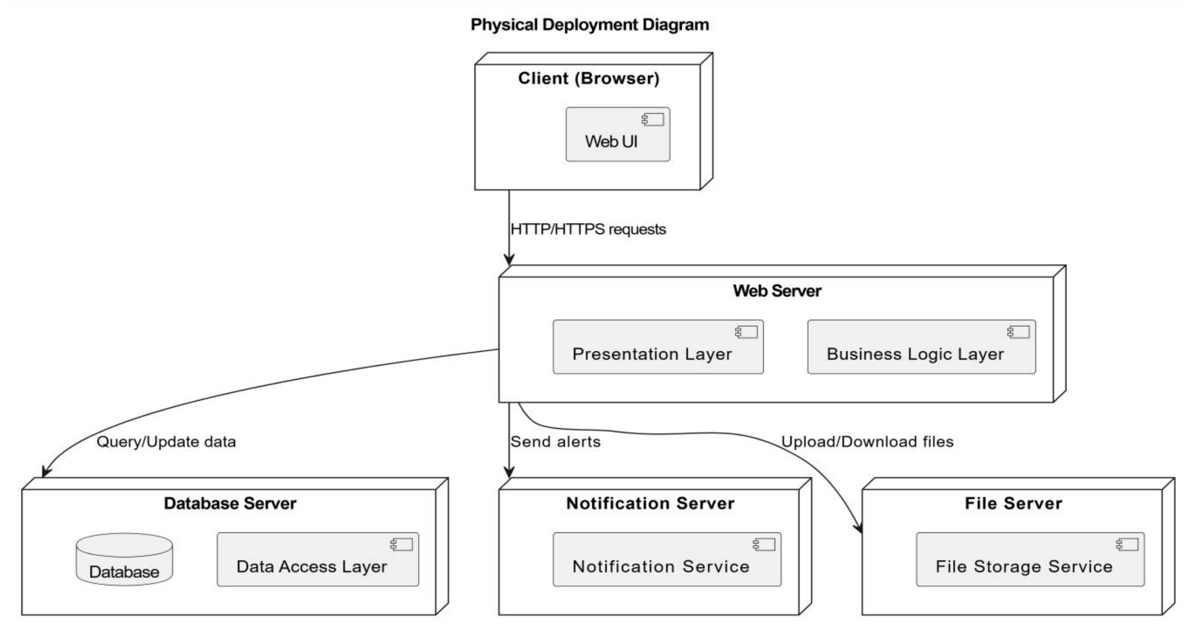

Figure 2.4.4 – Physical Deployment Diagram

CHAPTER THREE: DATA DESIGN

3.1 Logical Data Model

This chapter describes the data design of the Web-Based End-to-End SCM System. The purpose of this chapter is to translate the data requirements defined in the SRS into a structured data model that supports efficient storage, retrieval, integrity, and consistency of system data.

The logical data model is derived from the conceptual ERD defined in the SRS. It identifies the main entities, their attributes, primary keys, foreign keys, and relationships. The system follows a relational database approach where data is normalized to reduce redundancy and maintain data integrity.

3.1.1 Logical Tables and Attributes

User Table

- user_id (PK)
- email
- password_hash
- user_type (Administrator, Supplier, Retailer, Warehouse Manager, Delivery Agent, Car Rental Agency)
- supplier_id (FK → User.user_id, for Warehouse Managers only)
- company_name
- address
- phone
- business_license_number
- license_status
- registration_date
- account_status

PurchaseOrder Table

- po_id (PK)
- po_number (Unique)
- retailer_id (FK → User.user_id)
- supplier_id (FK → User.user_id)
- status (Draft, Issued, Accepted, Rejected, Converted)
- total_amount
- notes
- created_date

PurchaseOrderItem Table

- po_item_id (PK)
- po_id (FK → PurchaseOrder.po_id)
- product_id (FK → Product.product_id)
- quantity
- unit_price
- •subtotal

Product Table

- product_id (PK)
- supplier_id (FK → User.user_id)
- product_name
- description
- category
- unit_price
- quantity_available
- warehouse_location
- created_date

Order Table

- order_id (PK)
- po_id (FK → PurchaseOrder.po_id)
- retailer_id (FK → User.user_id)
- supplier_id (FK → User.user_id)
- order_date
- total_amount
- order_status
- delivery_address
- expected_delivery_date
- actual_delivery_date

OrderItem Table

- order_item_id (PK)
- order_id (FK → Order.order_id)
- product_id (FK → Product.product_id)
- supplier_id (FK → User.user_id)
- quantity
- unit_price
- subtotal

Delivery Table

- delivery_id (PK)
- order_id (FK → Order.order_id)
- tracking_number
- delivery_status
- estimated_delivery_date
- actual_delivery_date

- proof_of_delivery

VehicleRental Table

- rental_id (PK)
- supplier_id (FK → User.user_id)
- agency_id (FK → User.user_id)
- delivery_id (FK → Delivery.delivery_id)
- vehicle_details
- vehicle_type
- plate_number
- capacity
- rental_status (Requested, Approved, Rejected, Assigned, In Use, Returned, Completed)
- rental_start_date
- rental_end_date
- created_date

Rating Table

- rating_id (PK)
- supplier_id (FK → User.user_id)
- retailer_id (FK → User.user_id)
- order_id (FK → Order.order_id)
- product_quality_rating
- delivery_timeliness_rating
- communication_rating
- overall_rating
- feedback_comment

PaymentConfirmation Table

- confirmation_id (PK)

- order_id (FK → Order.order_id)
- confirmation_reference
- confirmation_date
- status

Commission Table

- commission_id (PK)
- order_id (FK → Order.order_id)
- supplier_id (FK → User.user_id)
- commission_rate
- commission_amount
- commission_status

Notification Table

- notification_id (PK)
- user_id (FK → User.user_id)
- role
- title
- message
- related_entity_id
- related_entity_type
- notification_status
- created_date

InventoryHistory Table

- inventory_history_id (PK)
- product_id (FK → Product.product_id)
- supplier_id (FK → User.user_id)
- warehouse_manager_id (FK → User.user_id)
- change_type

- quantity_changed
- change_reason
- change_date

AuditLog Table

- audit_id (PK)
- user_id (FK → User.user_id)
- action
- entity_type
- entity_id
- timestamp
- ip_address

3.1.2 Logical Relationships

- ✓ One Supplier → many Products (1:N)
- ✓ One Retailer → many PurchaseOrders (1:N)
- ✓ One PurchaseOrder → one or more Orders (1:N), supporting PO splitting with traceability
- ✓ One PurchaseOrder → multiple PurchaseOrderItems (1:N)
- ✓ One Retailer → many Orders (1:N)
- ✓ One Supplier → many Orders (1:N)
- ✓ Each Order → one Delivery (1:1)
- ✓ Each Delivery → optional VehicleRental (1:1)
- ✓ Completed Order → optional Rating (1:1)
- ✓ Each Order → one PaymentConfirmation (1:1) (records external proof only)
- ✓ Each Order → one Commission (1:1)
- ✓ One Supplier → multiple Commission records (1:N)
- ✓ One Order → multiple OrderItems (1:N)
- ✓ One Product → multiple OrderItems (1:N)
- ✓ One User → multiple Notifications (1:N)

- ✓ One Product → multiple InventoryHistory records (1:N)
- ✓ One VehicleRental → one Car Rental Agency (1:1)
- ✓ One Delivery Agent → multiple Deliveries (1:N)

This logical model ensures full traceability from procurement to delivery completion and commission recording, with optional vehicle rental coordination.

3.2 Physical Data Management

The system uses a **Relational Database Management System** as the primary data management approach. A relational database is selected due to its strong support for data integrity, transactional consistency, structured querying, and ease of maintenance.

3.2.1 Database Technology

- Database Type: Relational Database
- Suggested DBMS: MySQL or PostgreSQL
- Data Access Method: SQL

3.2.2 Physical Storage Strategy

- Data is stored in normalized tables with clearly defined primary keys and foreign key constraints.
- Indexes are applied on frequently queried attributes such as user email, product name, order status, PO number, and foreign keys.
- Referential integrity is enforced using foreign key constraints.
- Cascading rules are applied carefully to prevent accidental data loss.

3.2.3 Backup and Recovery

- Periodic database backups shall be performed to prevent data loss.
- Backup files shall be stored securely and retained for recovery purposes.
- Recovery procedures shall support restoration to a consistent state after system failure.

3.3 Data Dictionary

This section provides a detailed description of database tables, attributes, data types, and constraints.

3.3.1 User Table

Attribute	Data Type	Size	Description
user_id	INT	---	Primary key, unique identifier
email	VARCHAR	100	User login email (unique)
password_hash	VARCHAR	255	Encrypted password
user_type	VARCHAR	30	Administrator, Supplier, Retailer, Warehouse Manager, Delivery Agent, Car Rental Agency
supplier_id	INT	---	Foreign key to Supplier (for Warehouse Managers only)
company_name	VARCHAR	150	Registered business name
address	VARCHAR	255	Business address
phone	VARCHAR	20	Contact phone number
business_license_number	VARCHAR	50	Official license ID
license_status	VARCHAR	20	Pending, Approved, Rejected
registration_date	DATE	---	Account creation date
account_status	VARCHAR	20	Active, Suspended

3.3.2 PurchaseOrder Table

Attribute	Data Type	Size	Description

Attribute	Data Type	Size	Description
po_id	INT	---	Primary key
po_number	VARCHAR	50	Unique purchase order number
retailer_id	INT	---	Foreign key to User (Retailer)
supplier_id	INT	---	Foreign key to User (Supplier)
status	VARCHAR	30	Draft, Issued, Accepted, Rejected, Converted
total_amount	DECIMAL	12,2	Total PO value
notes	TEXT	---	Optional notes
created_date	DATE	---	PO creation date

3.3.3 PurchaseOrderItem Table

Attribute	Data Type	Size	Description
po_item_id	INT	—	Primary key; uniquely identifies each purchase order item
po_id	INT	—	References the related Purchase Order
product_id	INT	—	References the ordered product
quantity	INT	—	Quantity of the product requested
unit_price	DECIMAL	10,2	Price per unit of the product
subtotal	DECIMAL	12,2	Calculated as quantity × unit_price

3.3.4 Product Table

Attribute	Data Type	Size	Description
product_id	INT	---	Primary key
supplier_id	INT	---	Foreign key to User

Attribute	Data Type	Size	Description
product_name	VARCHAR	150	Product name
description	TEXT	---	Product description
category	VARCHAR	100	Product category
unit_price	DECIMAL	10,2	Product price
quantity_available	INT	---	Available stock
warehouse_location	VARCHAR	150	Storage location
created_date	DATE	---	Date added

3.3.5 Order Table

Attribute	Data Type	Size	Description
order_id	INT	---	Primary key
po_id	INT	---	Foreign key to PurchaseOrder (NEW)
retailer_id	INT	---	Retailer placing order
supplier_id	INT	---	Supplier fulfilling order
order_date	DATETIME	---	Order creation date
total_amount	DECIMAL	12,2	Total order value
order_status	VARCHAR	30	Pending, Approved, Shipped, Delivered, Completed
delivery_address	VARCHAR	255	Delivery location
expected_delivery_date	DATE	---	Planned delivery
actual_delivery_date	DATE	---	Actual delivery

3.3.6 OrderItem Table

Attribute	Data Type	Size	Description
order_item_id	INT	---	Primary key
order_id	INT	---	Related order
product_id	INT	---	Ordered product
supplier_id	INT	---	Product supplier
quantity	INT	---	Ordered quantity
unit_price	DECIMAL	10,2	Price per unit
subtotal	DECIMAL	12,2	quantity × unit_price

3.3.7 Delivery Table

Attribute	Data Type	Size	Description
delivery_id	INT	---	Primary key
order_id	INT	---	Associated order
tracking_number	VARCHAR	50	Delivery tracking ID
delivery_status	VARCHAR	30	Packed, In Transit, Delivered
estimated_delivery_date	DATE	---	Estimated date
actual_delivery_date	DATE	---	Actual date
proof_of_delivery	TEXT	---	Photo or signature reference

3.3.8 VehicleRental Table

Attribute	Data Type	Size	Description
rental_id	INT	---	Primary key
supplier_id	INT	---	Foreign key to User (Supplier)
agency_id	INT	---	Foreign key to User (Car Rental Agency)
delivery_id	INT	---	Foreign key to Delivery
vehicle_details	VARCHAR	200	Vehicle description
vehicle_type	VARCHAR	50	Truck, Van, Lorry, etc.
plate_number	VARCHAR	20	Vehicle license plate
capacity	VARCHAR	50	Load capacity
rental_status	VARCHAR	30	Requested, Approved, Rejected, Assigned, In Use, Returned, Completed
rental_start_date	DATE	---	Rental start date
rental_end_date	DATE	---	Rental end date
created_date	DATETIME	---	Creation timestamp

3.3.9 Rating Table

Attribute	Data Type	Size	Description
rating_id	INT	---	Primary key
supplier_id	INT	---	Rated supplier
retailer_id	INT	---	Rating retailer
order_id	INT	---	Related order
product_quality_rating	INT	---	Rating value (1-5)
delivery_timeliness_rating	INT	---	Rating value (1-5)

Attribute	Data Type	Size	Description
communication_rating	INT	---	Rating value (1-5)
overall_rating	DECIMAL	3,2	Average score
feedback_comment	TEXT	---	Optional comment

3.3.10 PaymentConfirmation Table

Attribute	Data Type	Size	Description
confirmation_id	INT	---	Primary key
order_id	INT	---	Related order
confirmation_reference	VARCHAR	100	External payment reference
confirmation_date	DATE	---	Confirmation date
status	VARCHAR	20	Pending, Confirmed
confirmed_by	INT(FK → User.user_id)	---	Retailer who confirmed payment
confirmation_method	VARCHAR(20)	20	Manual or Upload

3.3.11 Commission Table

Attribute	Data Type	Size	Description
commission_id	INT	---	Primary key
order_id	INT	---	Related order
supplier_id	INT	---	Supplier ID
commission_rate	DECIMAL	5,2	Percentage rate
commission_amount	DECIMAL	12,2	Calculated amount

Attribute	Data Type	Size	Description
commission_status	VARCHAR	20	Pending, Paid
verified_by_admin_id	INT (FK → User.user_id)	---	verified
verification_date	DATE	---	

3.3.12 Notification Table

Attribute	Data Type	Size	Description
notification_id	INT	---	Primary key
user_id	INT	---	Notification recipient
role	VARCHAR	20	Supplier, Retailer, Admin, Delivery Agent, Car Rental Agency
title	VARCHAR	150	Notification title
message	TEXT	---	Notification message
related_entity_id	INT	---	Related order, delivery, PO, or rental
related_entity_type	VARCHAR	50	Order, Delivery, PurchaseOrder, VehicleRental, Payment
notification_status	VARCHAR	20	Read, Unread
created_date	DATETIME	---	Creation timestamp

3.3.13 InventoryHistory Table

Attribute	Data Type	Size	Description
inventory_history_id	INT	---	Primary key

Attribute	Data Type	Size	Description
product_id	INT	---	Related product
supplier_id	INT	---	Supplier
change_type	VARCHAR	20	Addition, Reduction
quantity_changed	INT	---	Quantity modified
change_reason	VARCHAR	100	Order, restock, adjustment
change_date	DATETIME	---	Change timestamp

3.3.14 AuditLog Table

Attribute	Data Type	Size	Description
audit_id	INT	—	Primary key; uniquely identifies each audit record
user_id	INT	—	References the user who performed the action
action	VARCHAR	100	Action performed (e.g., Login, Approval, Update)
entity_type	VARCHAR	50	Type of affected entity (User, Order, PurchaseOrder)
entity_id	INT	—	Identifier of the affected entity
timestamp	DATETIME	—	Date and time when the action occurred
ip_address	VARCHAR	45	IP address of the user performing the action

This data design ensures consistency with the SRS, supports traceability of supply chain transactions including procurement and vehicle rental, and provides a solid foundation for database implementation and future system scalability.

CHAPTER FOUR: HUMAN INTERFACE DESIGN

4.1 Overview

This chapter describes the Human Interface Design of the Web-Based End-to-End SCM System. The purpose of this chapter is to explain how users interact with the system through graphical user interfaces and how the interface design supports usability, efficiency, and clarity.

The system is designed as a web-based application accessible through standard web browsers. The user interface follows a simple, intuitive, and role-based design to ensure that users with basic computer skills can easily perform their tasks. Each authenticated user role Administrator, Supplier, Retailer, Delivery Agent, and Car Rental Agency is provided with a dedicated dashboard that displays relevant information and actions.

The interface emphasizes:

- Clear navigation and consistent layout across pages
- Role-specific access to features
- Immediate feedback for user actions
- Error prevention and validation messages
- Simple workflows for core supply chain operations

From the user's perspective, the system allows seamless completion of tasks such as procurement, product management, order placement, vehicle rental coordination, delivery tracking, confirmation, and reporting, with appropriate feedback messages displayed at each step.

4.2 Screen Images

This section presents the major system screens from the user's perspective. The screen images may be implemented using wireframes, mockups, or hand-drawn illustrations. Each screen represents a key system function and reflects the actual layout that will be implemented during development.

The major screens included in the system are:

- Login Screen

- User Registration Screen
- Administrator Dashboard
- Supplier Dashboard
- Retailer Dashboard
- Warehouse Manager Dashboard
- PO Management Screen
- Product Management Screen
- Order Management Screen
- Vehicle Rental Management Screen
- Order Tracking Screen
- Payment Confirmation and Commission Screen

4.3 Screen Images and Description

This section provides a detailed description of each major screen, including its purpose, main interface components, and user interactions.

4.3.1 Login Screen

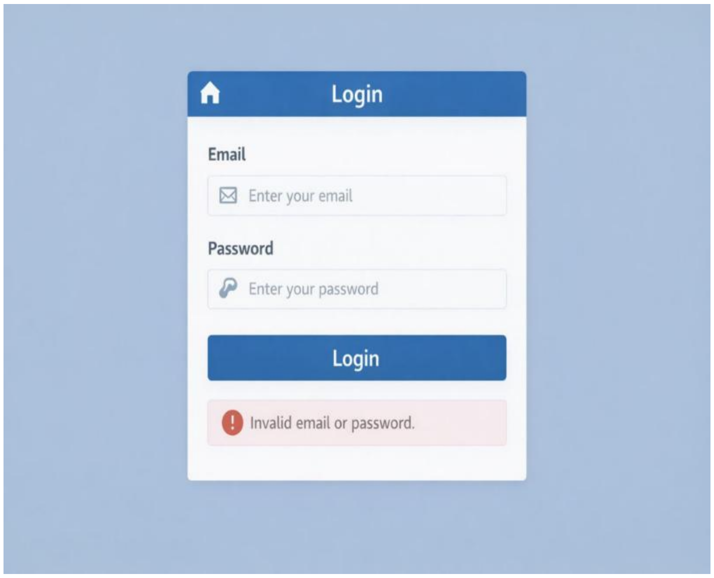

Image 4.1 Login Screen

Description:

The Login Screen allows registered users to securely access the system using their credentials.

Screen Element	Description
Email Field	Allows users to enter their registered email address
Password Field	Allows users to enter their password
Login Button	Authenticates user credentials
Error Message Area	Displays invalid login messages

4.3.2 User Registration Screen

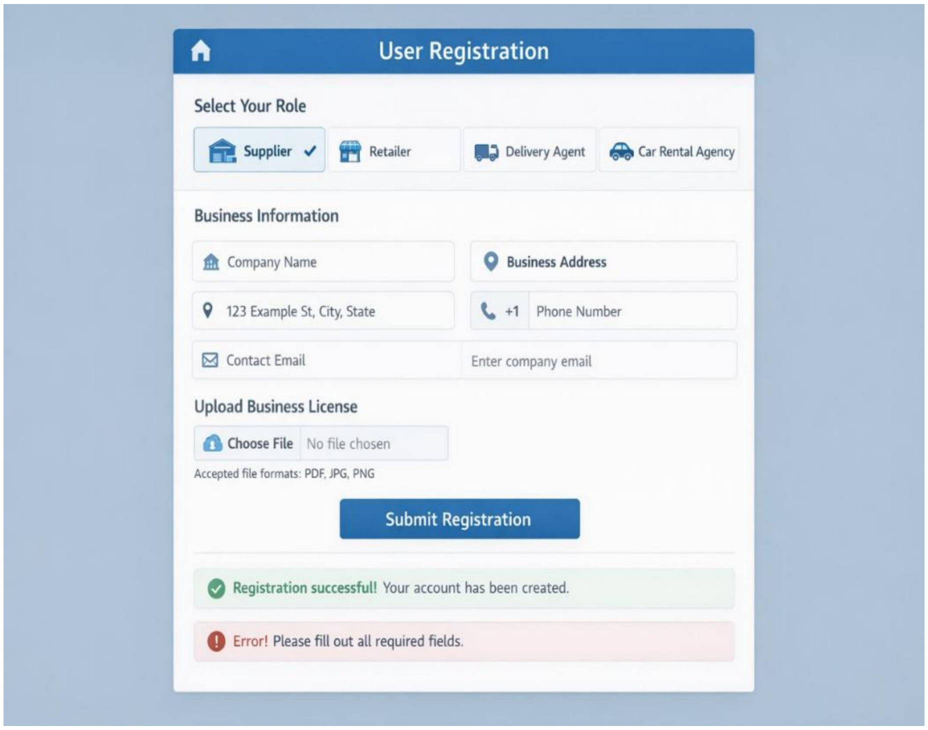

Image 4.2 User Registration Screen

Description:

This screen enables suppliers, retailers, delivery agents, and car rental agencies to create a new account by submitting personal and business information.

Screen Element	Description
Role Selection	Choose Supplier, Retailer, Delivery Agent, or Car Rental Agency
Business Information Form	Collects company name, address, and contact details
License Upload Field	Uploads business license document
Submit Button	Sends registration request for admin approval
Status Message	Displays registration success or error messages

4.3.3 Administrator Dashboard

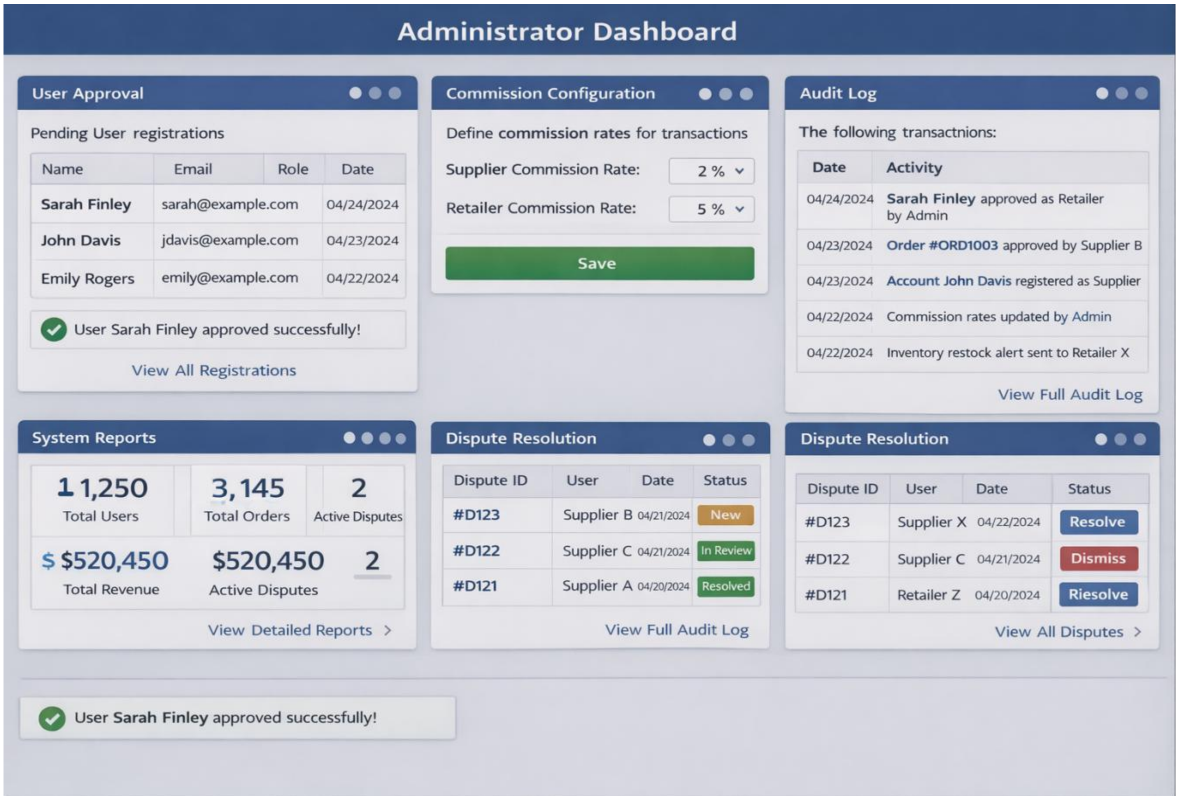

Image 4.3 Administrator Dashboard

Description:

The Administrator Dashboard provides system oversight and management tools without interfering in supplier--retailer transactions.

Screen Element	Description
User Approval Panel	Approves or rejects all user registrations
Commission Configuration	Sets commission rates for the platform
System Reports	Displays system-wide statistics and analytics
Audit Log Viewer	Shows logged system activities and transactions. Displays read-only records retrieved from the system AuditLog data store for monitoring and accountability purposes.
Dispute Resolution	Handles user-reported disputes

Screen Element	Description
Panel	

4.3.4 Supplier Dashboard

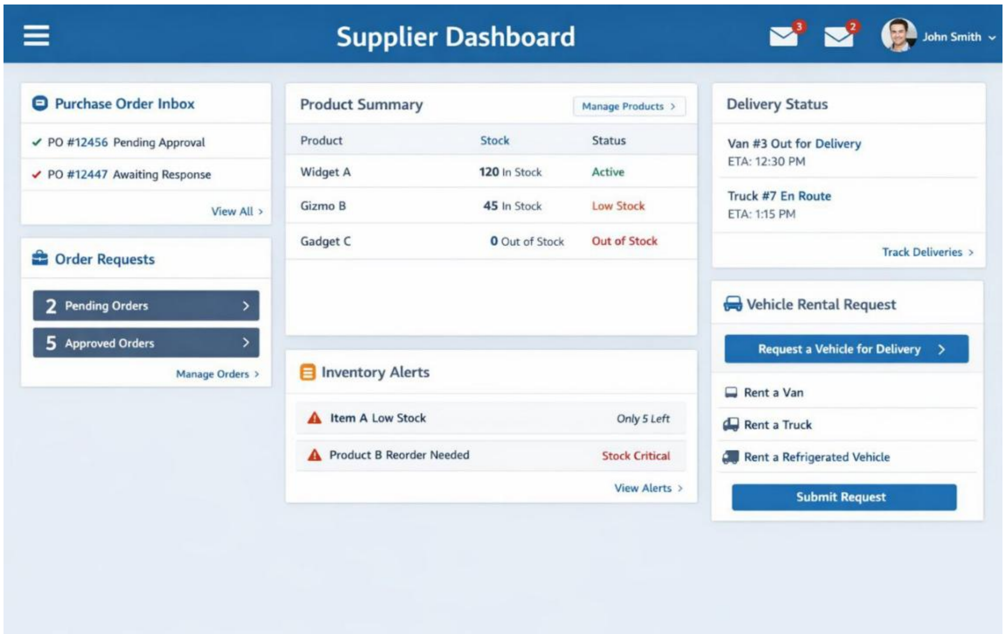

Image 4.4 Supplier Dashboard

Description:

The Supplier Dashboard allows suppliers to manage products, inventory, POs, orders, and delivery operations.

Screen Element	Description
Purchase Order Inbox	Shows pending POs requiring acceptance/rejection
Product Summary	Displays listed products and stock levels
Order Requests	Shows pending and approved orders
Inventory Alerts	Displays low-stock notifications
Delivery Status Panel	Shows ongoing deliveries
Vehicle Rental Request Panel	Allows requesting vehicles for deliveries

4.3.5 Retailer Dashboard

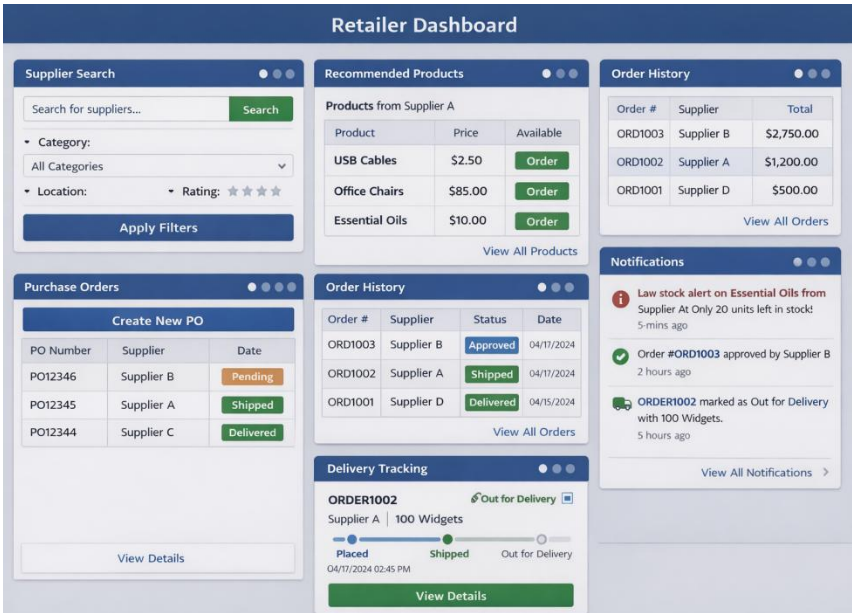

Image 4.5 Retailer Dashboard

Description:

The Retailer Dashboard provides tools for product discovery, procurement, order placement, and order tracking.

Screen Element	Description
Supplier Search Bar	Search and filter suppliers
Purchase Order Management	Create and manage POs
Order History	Displays past and current orders
Delivery Tracking	Shows real-time delivery status
Notifications	Displays system alerts and updates

4.3.6 Warehouse Manager Dashboard

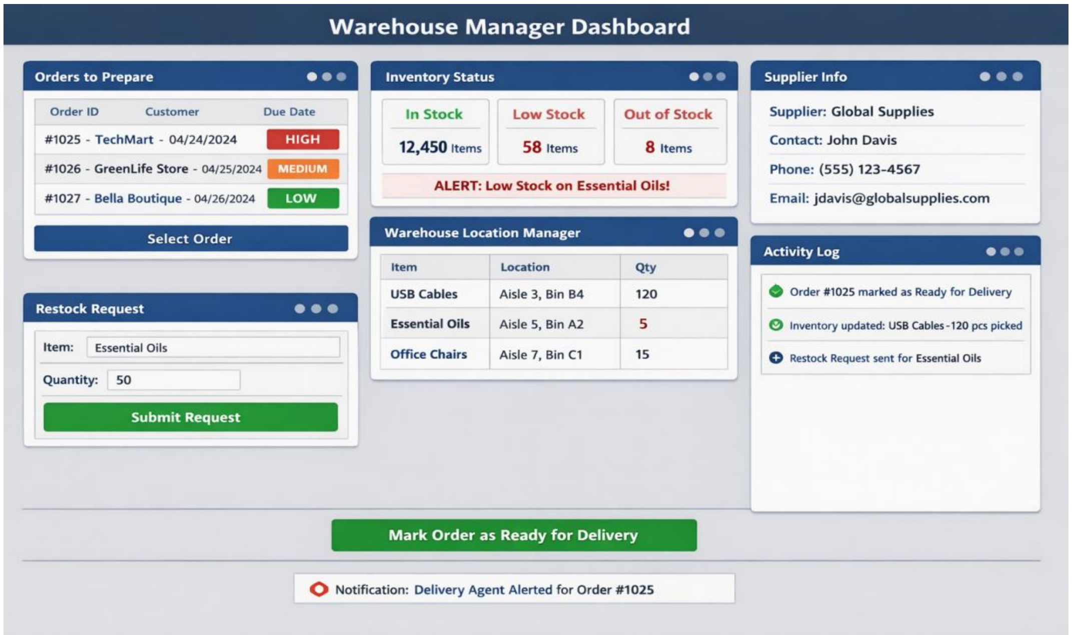

Image 4.6 Warehouse Manager Dashboard

Description:

The Warehouse Manager Dashboard provides tools for inventory management, order preparation, and warehouse operations under supplier authority.

This dashboard operates as a supplier sub-role and does not represent an independent system role.

Screen Element	**Description**
Orders to Prepare Panel	Displays orders requiring picking and packing with priority indicators
Inventory Status Dashboard	Shows current stock levels with low-stock alerts
Warehouse Location Manager	Manages item storage locations and bin assignments
Restock Request Form	Submits requests to supplier for low stock items
Activity Log Viewer	Shows recent warehouse operations and changes
Supplier Information Panel	Displays assigned supplier details and contact information

User Interactions:

1. Warehouse Manager logs in with credentials
2. System displays pending orders for preparation
3. Manager selects order for processing
4. System shows item locations and quantities
5. Manager picks items and updates system
6. System automatically deducts from inventory
7. Manager marks order as ready for delivery
8. System notifies Delivery Agent

4.3.7 Purchase Order Management Screen

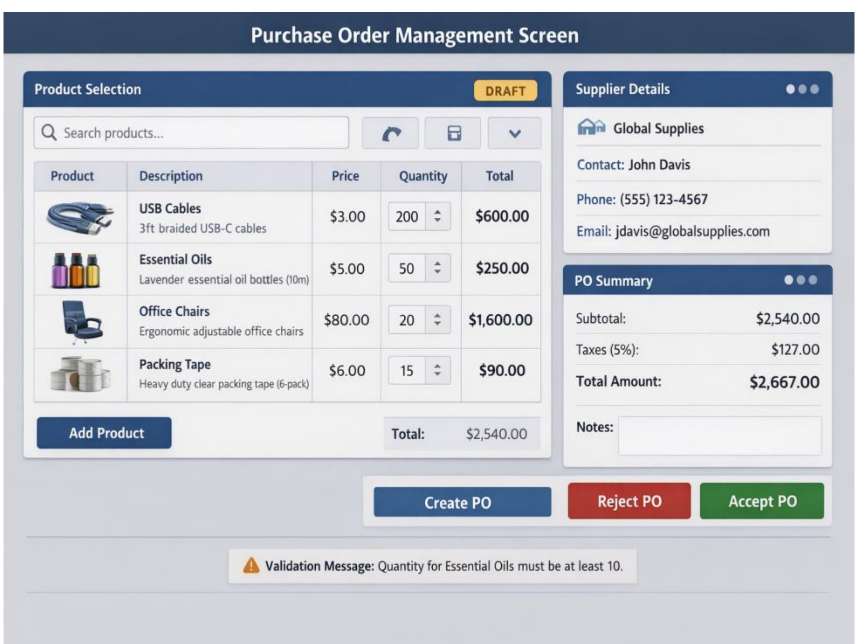

Image 4.7 Purchase Order Management Screen

Description:

This screen enables retailers to create formal POs and allows suppliers to review and accept/reject POs before order creation. The "Converted" status indicates that an accepted PO has been transformed into one or more system Orders.

Screen Element	**Description**
Product Selection Panel	Select products and quantities for PO
Supplier Details Display	Shows selected supplier information
PO Summary Section	Displays calculated total amount and details
Status Indicator	Shows PO status (Draft, Issued, Accepted, Rejected, Converted)
Action Buttons	Create PO (Retailer) / Accept/Reject (Supplier)
Validation Messages	Displays input errors or business rule violations

4.3.8 Product Management Screen

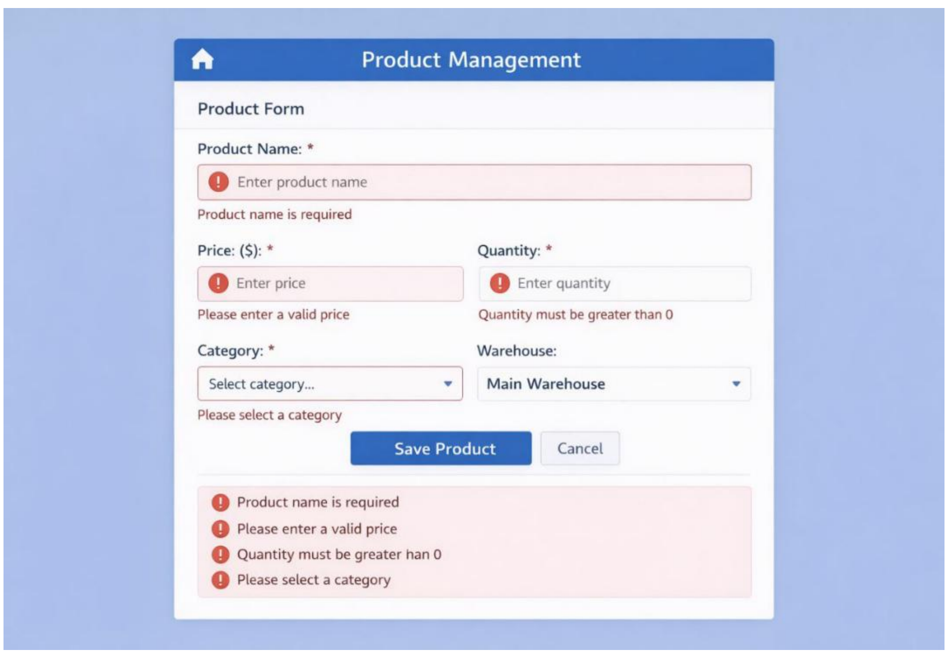

Image 4.8 Product Management Screen

Description:

This screen allows suppliers to add, update, or remove product listings and manage inventory quantities.

Screen Element	**Description**
Product Form	Inputs product name, price, quantity, and category
Warehouse Selector	Selects warehouse location
Save Button	Saves or updates product information
Validation Messages	Displays input errors

4.3.9 Order Management Screen

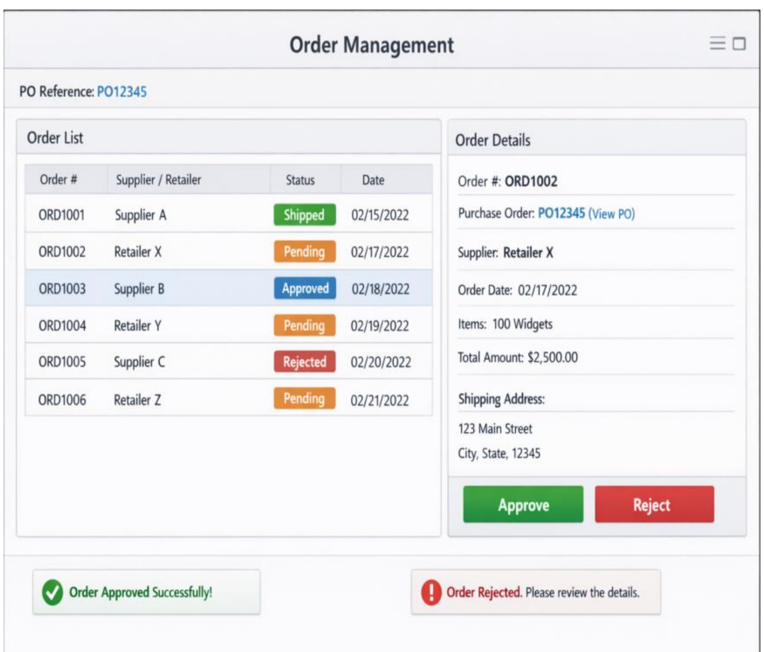

Image 4.9 Order Management Screen

Description:

This screen manages the order lifecycle for both suppliers and retailers, now referencing the originating PO.

Screen Element	Description
PO Reference Display	Shows related PO number
Order List	Displays orders with status indicators
Order Details Panel	Shows detailed order information including PO link
Approve/Reject Buttons	Allows supplier to process orders
Status Update Messages	Confirms action results

4.3.10 Vehicle Rental Management Screen

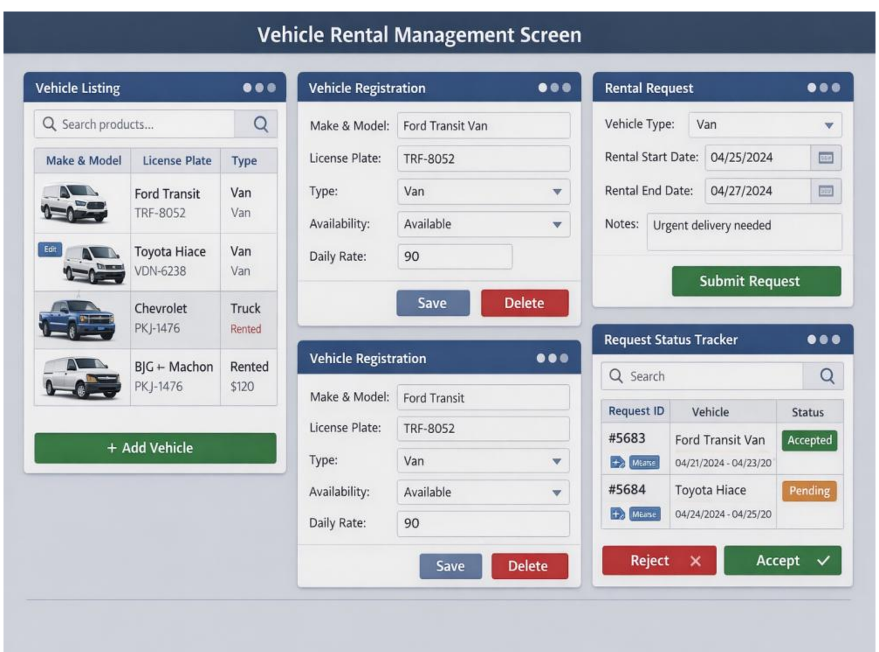

Image 4.10 Vehicle Rental Management Screen

Description:

This screen allows car rental agencies to list vehicles and manage availability, and enables suppliers to request vehicle rentals for deliveries.

Screen Element	**Description**
Vehicle Listing Panel	Displays available vehicles with details (Agency)
Vehicle Registration Form	Add/edit vehicle information (Agency)
Rental Request Panel	Submit rental requests for deliveries (Supplier)
Request Status Tracker	Shows status of rental requests
Action Buttons	Accept/Reject requests (Agency) / Submit Request (Supplier

4.3.11 Order Tracking Screen

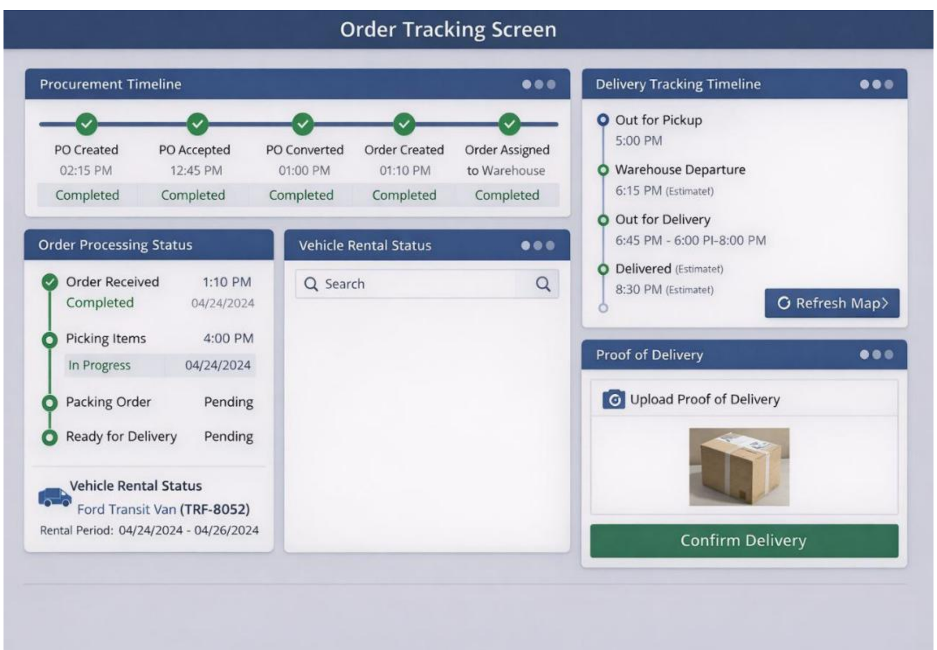

Image 4.11 Order Tracking Screen

Description:

The Order Tracking Screen provides visibility into the complete order lifecycle from PO to final delivery, including vehicle rental status when applicable.

Screen Element	Description
Procurement Timeline	Shows PO creation, acceptance, and conversion
Order Processing Status	Displays order fulfillment stages
Delivery Tracking Timeline	Shows delivery stages with estimated/actual times
Vehicle Rental Status	Displays rental request and assignment details
Proof Upload Field	Uploads photo or signature for delivery confirmation
Confirmation Button	Confirms successful delivery

4.3.12 Delivery Tracking Screen

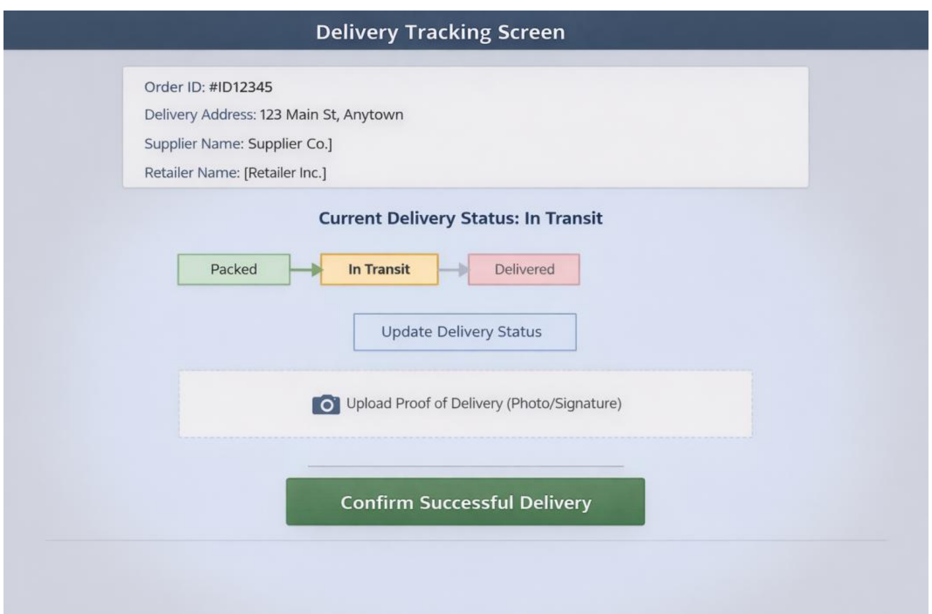

Image 4.12 – Delivery Tracking Screen

Description:

The Delivery Tracking Screen enables delivery agents, suppliers, and retailers to monitor the real-time status of an order from warehouse dispatch to final delivery. The screen provides clear visibility of delivery progress, key order details, and delivery confirmation actions, including proof of delivery upload and status updates.

Screen Element	**Description**
Order Information Panel	Displays Order ID, delivery address, supplier name, and retailer name
Current Delivery Status Indicator	Shows the current delivery state (Packed, In Transit, Delivered)
Delivery Progress Flow	Visual flow illustrating delivery stages and progression
Update Delivery Status Button	Allows the delivery agent to update the delivery stage
Proof of Delivery Upload Field	Enables upload of delivery proof (photo or customer signature)
Confirmation Button	Confirms successful delivery and finalizes the delivery process

User Interactions:

1. Delivery Agent logs into the system
2. System displays assigned delivery order details
3. Current delivery status is shown as Packed **or** In Transit
4. Delivery Agent updates delivery status as the order progresses
5. System reflects status changes in real time
6. Upon delivery, the agent uploads proof of delivery (photo or signature)
7. Delivery Agent confirms successful delivery
8. System marks the order as Delivered and updates order records

4.3.13 Payment Confirmation and Commission Screen

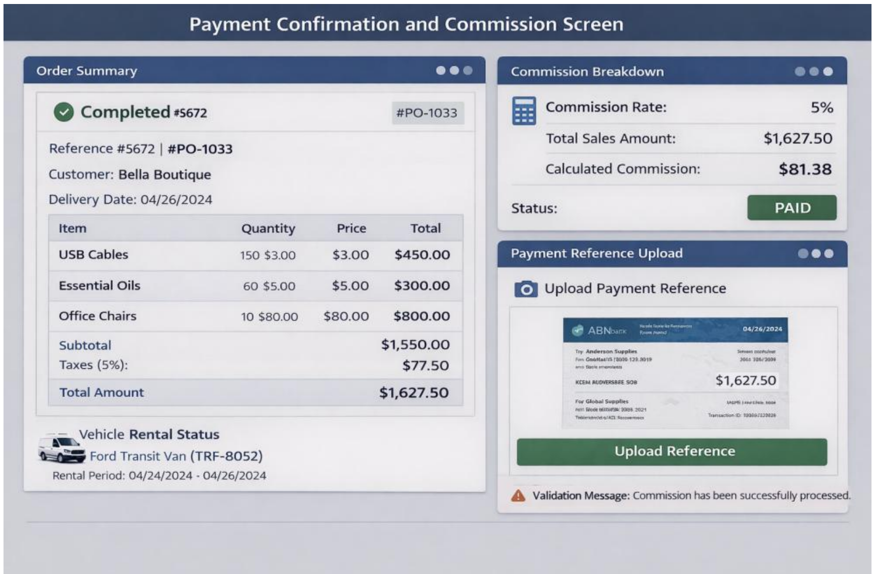

Image 4.13 Payment Confirmation and Commission Screen

Description:

This screen displays payment confirmation records and commission details for completed orders.

Screen Element	Description
Order Summary	Shows completed order details with PO reference
Commission Breakdown	Displays commission rate and calculated amount
Payment Reference Upload	Uploads external payment proof
Status Indicator	Shows commission payment status

The human interface design described in this chapter ensures that the system remains user-friendly, role-focused, and aligned with the functional and NFR defined in the SRS. The addition of PO and Vehicle Rental screens, along with the enhanced Order Tracking screen, provides a comprehensive interface for managing the complete supply chain workflow.

CHAPTER FIVE: REQUIREMENT MATRIX

5.1 Overview

This chapter presents the Requirement Matrix for the Web-Based End-to-End SCM System. The purpose of the requirement matrix is to establish clear traceability between the functional requirements specified in the SRS and the corresponding system components and design modules described in the SDD.

The requirement matrix ensures that:

- All functional requirements defined in the SRS are addressed by the system design
- Each system component has a clear purpose linked to one or more requirements

- The design can be verified, validated, and assessed during implementation and evaluation

This traceability is essential for academic assessment, system verification, and future maintenance.

5.2 Requirement Traceability Matrix

No.	SRS Requirement ID	Requirement Description	Design Element / Module
1	FR-3.1.1.1	User registration with business information	User Management Subsystem
2	FR-3.1.1.2	Business license upload and validation for all business users	User Management Subsystem
3	FR-3.1.1.3	Administrator approval of users	Administrative Oversight Subsystem
4	FR-3.1.1.4	Role-based access control (RBAC)	Authentication & Authorization Module
5	FR-3.1.1.5	Warehouse Managers operate under supplier authority	User Management Subsystem
6	FR-3.1.1.6	Delivery Agents independently authenticated and restricted	User Management Subsystem
7	FR-3.1.1.7	User authentication using username and password	Authentication & Authorization Module
8	FR-3.1.1.8	Audit logs for registration, approval, and authentication	Logging & Audit Module
9	FR-3.1.2.1	Supplier and product search	Supplier Discovery & Search Subsystem
10	FR-3.1.2.2	Supplier filtering by criteria	Supplier Discovery & Search Subsystem
11	FR-3.1.2.3	Save favorite suppliers for quick access	Supplier Discovery & Search Subsystem
12	FR-3.1.3.1	Product listing management	Product & Catalog Management Subsystem
13	FR-3.1.3.2	Product attributes (name, description, category, price, quantity, location)	Product & Catalog Management Subsystem
14	FR-3.1.3.3	Product–warehouse association	Inventory & Warehouse Management Subsystem
15	FR-3.1.4.1	Retailer PO creation	Procurement & PO

No.	SRS Requirement ID	Requirement Description	Design Element / Module
			Management Subsystem
16	FR-3.1.4.2	PO attributes (number, supplier, retailer, items, totals, delivery date)	Procurement & PO Management Subsystem
17	FR-3.1.4.3	"PO Issued" status for new POs	Procurement & PO Management Subsystem
18	FR-3.1.4.4	Supplier accepts/rejects PO with justification	Procurement & PO Management Subsystem
19	FR-3.1.4.5	Only accepted POs converted to Orders; one PO may generate multiple Orders if split	Order Processing Subsystem
20	FR-3.1.4.6	Complete PO history for auditing	Reporting & Analytics Subsystem
21	FR-3.1.5.1	Generate Order from approved PO	Order Processing Subsystem
22	FR-3.1.5.2	"Order Created" status for new Orders	Order Processing Subsystem
23	FR-3.1.5.3	Complete Order history for auditing	Reporting & Analytics Subsystem
24	FR-3.1.5.4	Prevent admin interference in transactions	Administrative Oversight Subsystem
25	FR-3.1.6.1	Suppliers/Warehouse Managers manage inventory levels	Inventory & Warehouse Management Subsystem
26	FR-3.1.6.2	Prevent negative inventory	Inventory & Warehouse Management Subsystem
27	FR-3.1.6.3	Inventory updates automatically upon Order generation	Inventory & Warehouse Management Subsystem
28	FR-3.1.7.1	Suppliers assign delivery tasks to Delivery Agents	Delivery & Logistics Management Subsystem
29	FR-3.1.7.2	Delivery Agents update order status	Delivery & Logistics Management Subsystem
30	FR-3.1.7.3	Support delivery using supplier-owned or rented vehicles	Delivery & Logistics Management Subsystem
31	FR-3.1.7.4	Delivery Agents cannot access pricing, inventory, or commission data	User Management Subsystem
32	FR-3.1.8.1	Retailer order tracking	Delivery & Logistics Management Subsystem
33	FR-3.1.8.2	Retailer confirmation before order completion	Delivery & Logistics Management Subsystem

No.	SRS Requirement ID	Requirement Description	Design Element / Module
34	FR-3.1.8.3	Proof of delivery using photo/signature	Delivery Confirmation Module
35	FR-3.1.8.4	QR code delivery confirmation (Optional)	Delivery & Logistics Management Subsystem
36	FR-3.1.9.1	Car Rental Agencies register vehicles	Vehicle Rental Management Subsystem
37	FR-3.1.9.2	Suppliers request vehicle rentals	Vehicle Rental Management Subsystem
38	FR-3.1.9.3	Car Rental Agencies approve/reject rental requests	Vehicle Rental Management Subsystem
39	FR-3.1.9.4	Assign approved vehicles to Delivery Agents	Vehicle Rental Management Subsystem
40	FR-3.1.10.1	Mark orders completed after retailer confirmation	Order Processing Subsystem
41	FR-3.1.10.2	Calculate administrative commission for suppliers (per completed PO)	Commission Management Subsystem
42	FR-3.1.10.3	Suppliers upload commission payment proof	Commission Management Subsystem
43	FR-3.1.10.4	Administrators verify/record commission settlement	Administrative Oversight Subsystem
44	FR-3.1.10.5	System does not process real financial transactions	Commission Management Subsystem
45	FR-3.1.11.1	Generate notifications for key events	Notification & Alert Management Subsystem
46	FR-3.1.11.2	Store and display notifications on dashboards	Notification & Alert Management Subsystem

5.3 Non-Functional Requirement Traceability Matrix

No.	SRS NFR ID	NFR Description	Design Element / Implementation
1	NFR-3.4.1.1	Load main dashboard within 7 seconds	Performance Optimization, Caching Layer
2	NFR-	Product search results within 5	Search Indexing, Database Optimization

No.	SRS NFR ID	NFR Description	Design Element / Implementation
	3.4.1.2	seconds	
3	NFR-3.4.1.3	PO creation/approval within 7 seconds	Business Logic Layer Optimization
4	NFR-3.4.1.4	Standard reports within 15 seconds	Reporting Engine, Query Optimization
5	NFR-3.4.1.5	Inventory queries within 5 seconds	Database Indexing, Caching
6	NFR-3.4.1.6	Support 200 concurrent users	Load Balancing, Connection Pooling
7	NFR-3.4.1.7	Process 50 orders/POs per hour	Transaction Processing Module
8	NFR-3.4.1.8	20 transactions per second	Database Optimization, Connection Management
9	NFR-3.4.1.9	CPU < 75% under normal load	Resource Management, Code Optimization
10	NFR-3.4.1.10	Memory < 80%	Memory Management, Garbage Collection
11	NFR-3.4.1.11	Network utilization minimal	Data Compression, Efficient APIs
12	NFR-3.4.1.12	Support future scaling	Modular Architecture, Microservices-ready
13	NFR-3.4.1.13	Support 5,000 users, 50,000 products	Scalable Database Design
14	NFR-3.4.1.14	Retain data for 3 years	Archival Strategy, Backup System
15	NFR-3.4.2.1	New retailer creates PO within 15 minutes	Intuitive UI, Guided Workflows
16	NFR-3.4.2.2	Supplier adds product within 20 minutes	Simplified Product Management UI
17	NFR-3.4.2.3	Common tasks in 5–6 interactions	Efficient Navigation, Shortcuts
18	NFR-3.4.2.4	Clear descriptive error messages	Validation Framework, User Feedback
19	NFR-3.4.2.5	Confirmation before delete/cancel	Confirmation Dialogs, Undo Features
20	NFR-3.4.2.6	Auto-save for long forms	Form State Management, Auto-save Service
21	NFR-	Basic accessibility practices	WCAG Compliance, Keyboard

No.	SRS NFR ID	NFR Description	Design Element / Implementation
	3.4.2.7		Navigation
22	NFR-3.4.2.8	Text resizing, color contrast	Responsive Design, Accessibility Features
23	NFR-3.4.3.1	Passwords securely hashed	Encryption Module, Secure Storage
24	NFR-3.4.3.2	Authentication before access	Authentication Service, Session Management
25	NFR-3.4.3.3	Role-based access control	RBAC Implementation, Permission System
26	NFR-3.4.3.4	Session expiration after inactivity	Session Timeout, Auto-logout
27	NFR-3.4.3.5	HTTPS for all communication	SSL/TLS Implementation, Secure Protocols
28	NFR-3.4.3.6	Protect sensitive data	Data Encryption, Access Controls
29	NFR-3.4.3.7	Restrict uploaded files	File Validation, Virus Scanning
30	NFR-3.4.3.8	Log significant system actions	Audit Logging System
31	NFR-3.4.3.9	Retain logs for troubleshooting	Log Management, Archival
32	NFR-3.4.4.1	Available during business hours	High Availability Design, Monitoring
33	NFR-3.4.4.2	Graceful failure handling	Error Recovery, Transaction Rollback
34	NFR-3.4.4.3	Backup and recovery procedures	Backup System, Disaster Recovery
35	NFR-3.4.5.1	Operate on modern browsers	Cross-browser Compatibility Testing
36	NFR-3.4.5.2	Usable on desktop and mobile	Responsive Web Design, Mobile-first
37	NFR-3.4.5.3	Deployable on common servers	Platform-independent Architecture
38	NFR-3.4.6.1	Comply with Ethiopian business regulations	Compliance Module, Regulatory Checks
39	NFR-3.4.6.2	Responsible user data handling	Data Privacy, GDPR-like Compliance
40	NFR-	Promote fair/ethical business	Business Rules Engine, Fairness

No.	SRS NFR ID	NFR Description	Design Element / Implementation
	3.4.6.3	practices	Controls

5.4 Business Rule Traceability Matrix

No.	SRS Business Rule ID	Business Rule Description	Enforced By
1	BR-01	Suppliers/retailers approved by admin before access	User Management Subsystem
2	BR-02	Products requiring regulatory approval need valid licenses	Product Validation Module
3	BR-03	Suppliers provide justification for PO rejection	Procurement & PO Management Subsystem
4	BR-04	Inventory never below zero	Inventory & Warehouse Management Subsystem
5	BR-05	Only accepted POs converted to Orders	Order Processing Subsystem
6	BR-06	Delivery tasks accepted before goods release	Delivery & Logistics Management Subsystem
7	BR-07	Proof of delivery mandatory before order completion	Delivery Confirmation Module
8	BR-08	Commission confirmation after PO completion (verified by Admin)	Commission Management Subsystem
9	BR-09	Commission applies only to completed supplier orders	Commission Management Subsystem
10	BR-10	Commission rates configurable only by administrators	Administrative Oversight Subsystem
11	BR-11	Suppliers receive net payment details after commission	Reporting & Analytics Subsystem
12	BR-12	Commission actions logged for audit	Logging & Audit Module
13	BR-13	Vehicle rentals approved by Car Rental Agency	Vehicle Rental Management Subsystem
14	BR-14	Critical system actions generate notifications	Notification & Alert Management Subsystem
15	BR-15	Notifications stored for audit	Notification & Alert Management Subsystem
16	BR-16	Users can view their notifications	Notification & Alert Management Subsystem

5.5 Requirement Coverage Summary

This Software Design Document provides comprehensive traceability between the SRS requirements and the system design components. The requirement matrices demonstrate that:

1. Functional Requirement Coverage

- All 46 functional requirements specified in SRS Section 3.1 are mapped to appropriate system subsystems and modules.
- Key capabilities covered:

  - ✧ **Procurement workflow** via Procurement & PO Management Subsystem
  - ✧ **Vehicle rental coordination** via Vehicle Rental Management Subsystem
  - ✧ **Enhanced role management** for all six user roles: Administrator, Supplier, Retailer, Warehouse Manager, Delivery Agent, Car Rental Agency
  - ✧ **Complete order lifecycle tracking**: PO creation → Order generation → Delivery confirmation
  - ✧ **Security controls**: RBAC, authentication, and data protection
  - ✧ **Performance optimization**: Caching, indexing, and efficient architecture
  - ✧ **Usability features**: guided workflows, accessibility compliance

2. Non-Functional Requirement Coverage

- ✧ All 40 non-functional requirements from SRS Section 3.4 are addressed through specific design components and implementation strategies.

3. Business Rule Enforcement

- ✧ All 16 business rules from SRS Section 2.4 are enforced by designated system components.

4. Design Validation

- ✧ No critical requirements have been omitted during the design phase.
- ✧ Each SRS requirement has a corresponding design element that implements or supports it.

5. Traceability Support

The traceability matrices provide clear references for:

- ✧ Academic assessment and verification
- ✧ System implementation guidance
- ✧ Future enhancement planning
- ✧ Maintenance and troubleshooting

This comprehensive requirement traceability ensures that the system design fully satisfies all specified requirements and provides a solid foundation for implementation, testing, and evaluation**.**